بسم الله الرحمن الرحيم



السلام عليكم ورحمة الله وبركاته

Anemia





By

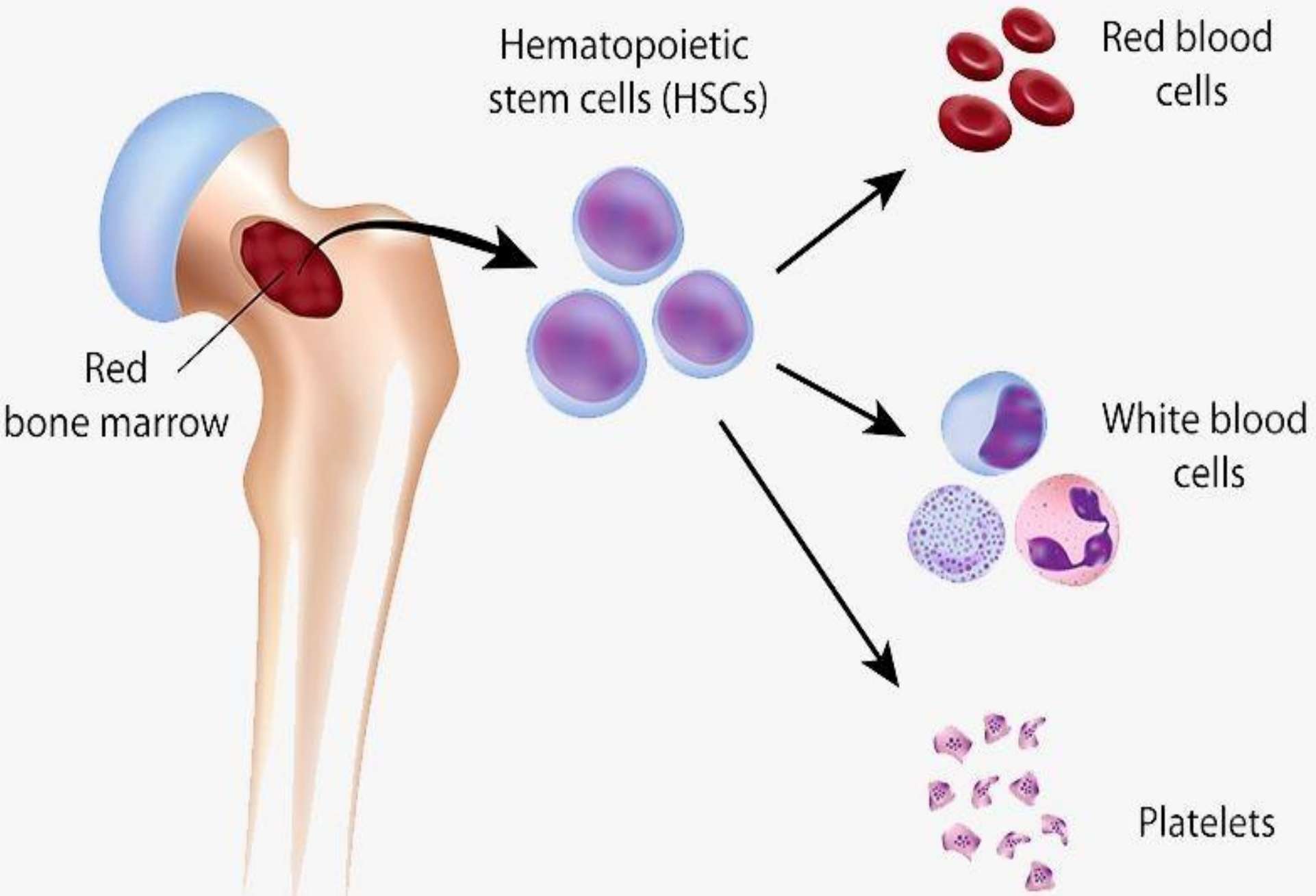
Eman Fathala Gad .

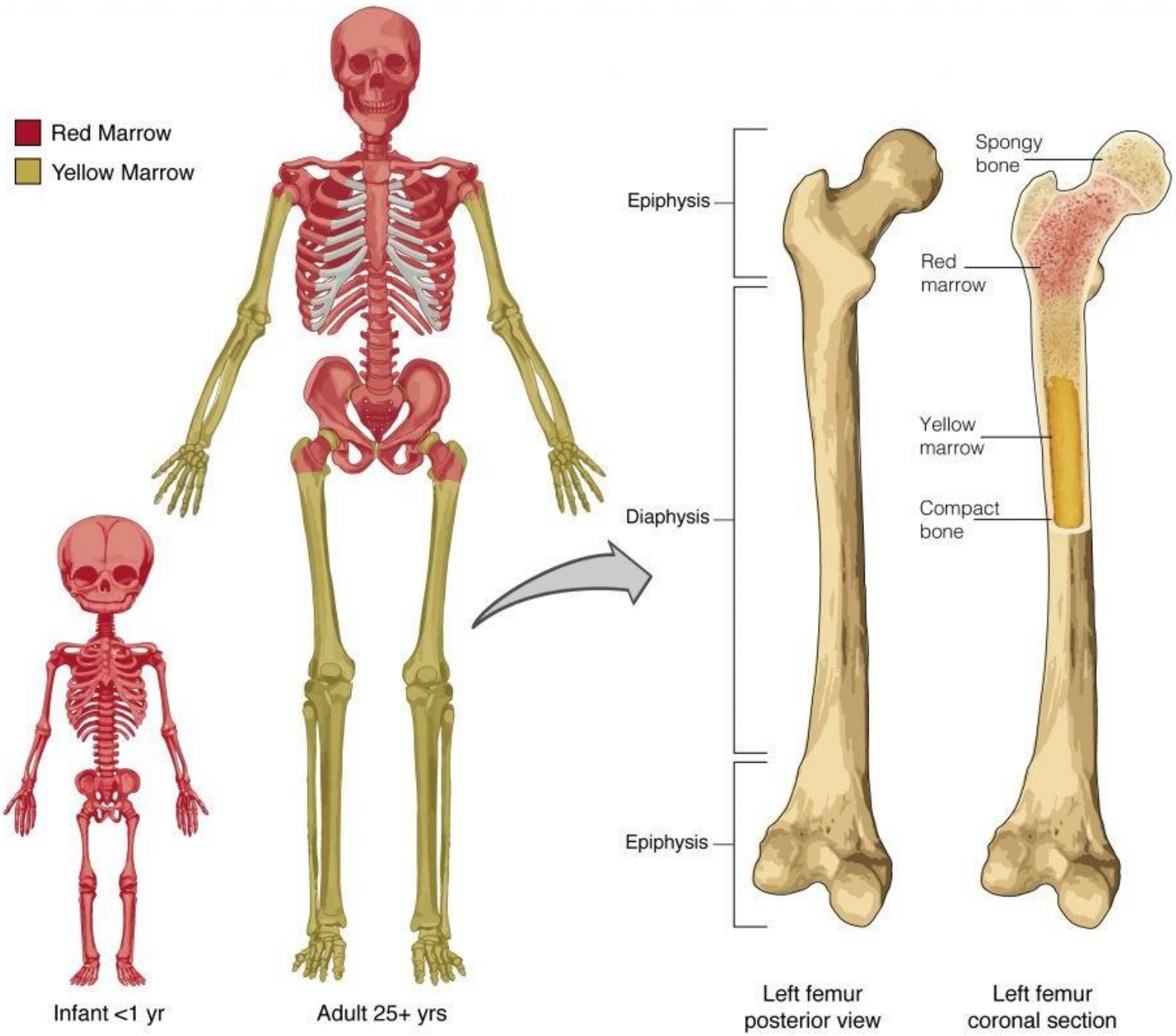
**Assistant Prof. of Pediatrics – Assiut University*

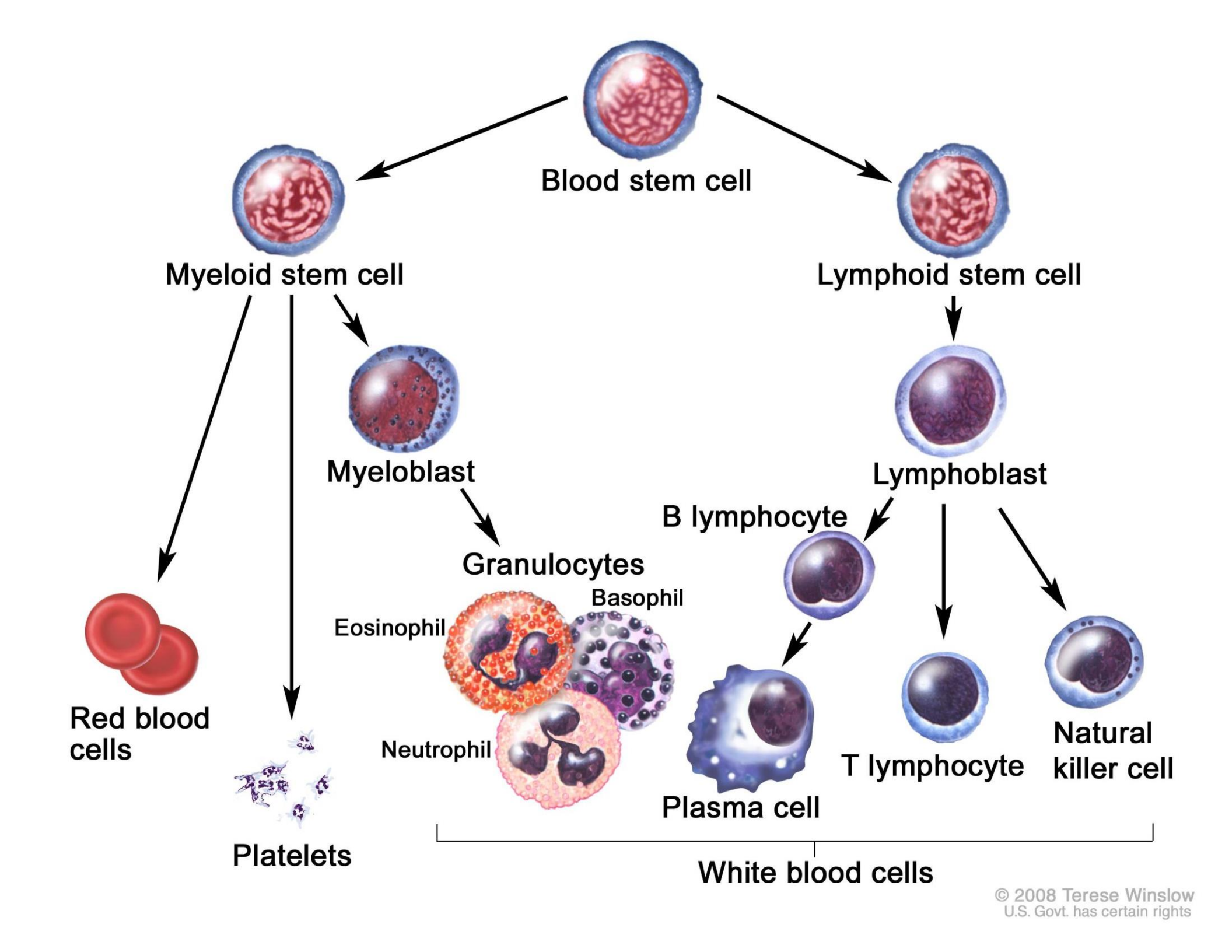






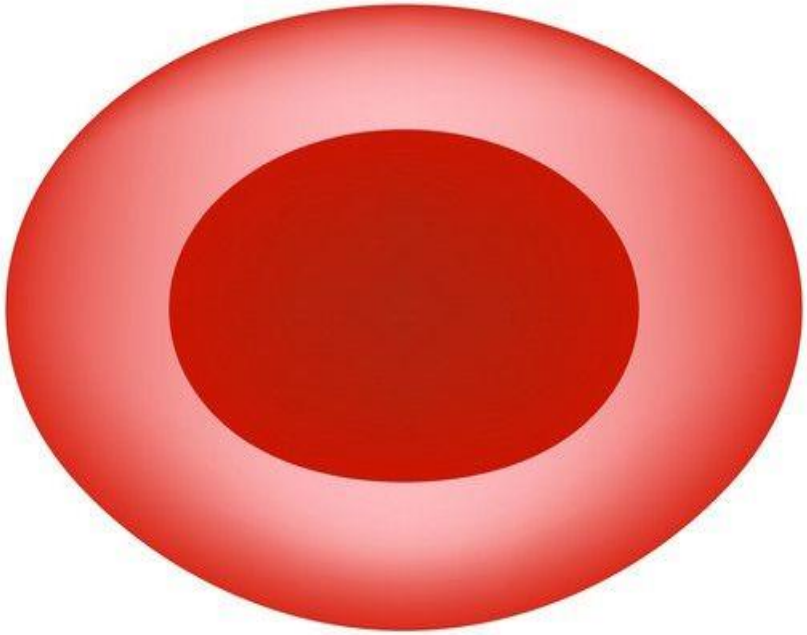

Hematopoietic
stem cells (HSCs)
Red blood
cells
Red
bone marrow
White blood
cells
Platelets

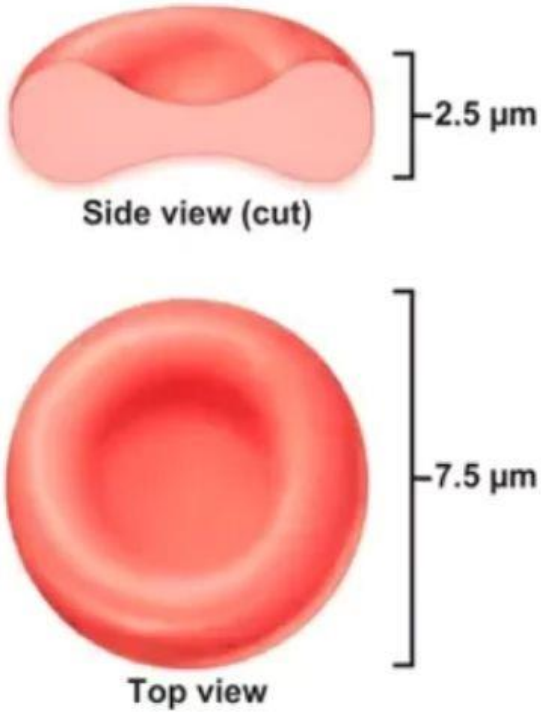

Red Marrow
Yellow Marrow
Epiphysis
Diaphysis
Epiphysis
Spongy bone
Red marrow
Yellow marrow
Compact bone
Infant <1 yr
Adult 25+ yrs
Left femur posterior view
Left femur coronal section



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Erythrocyte

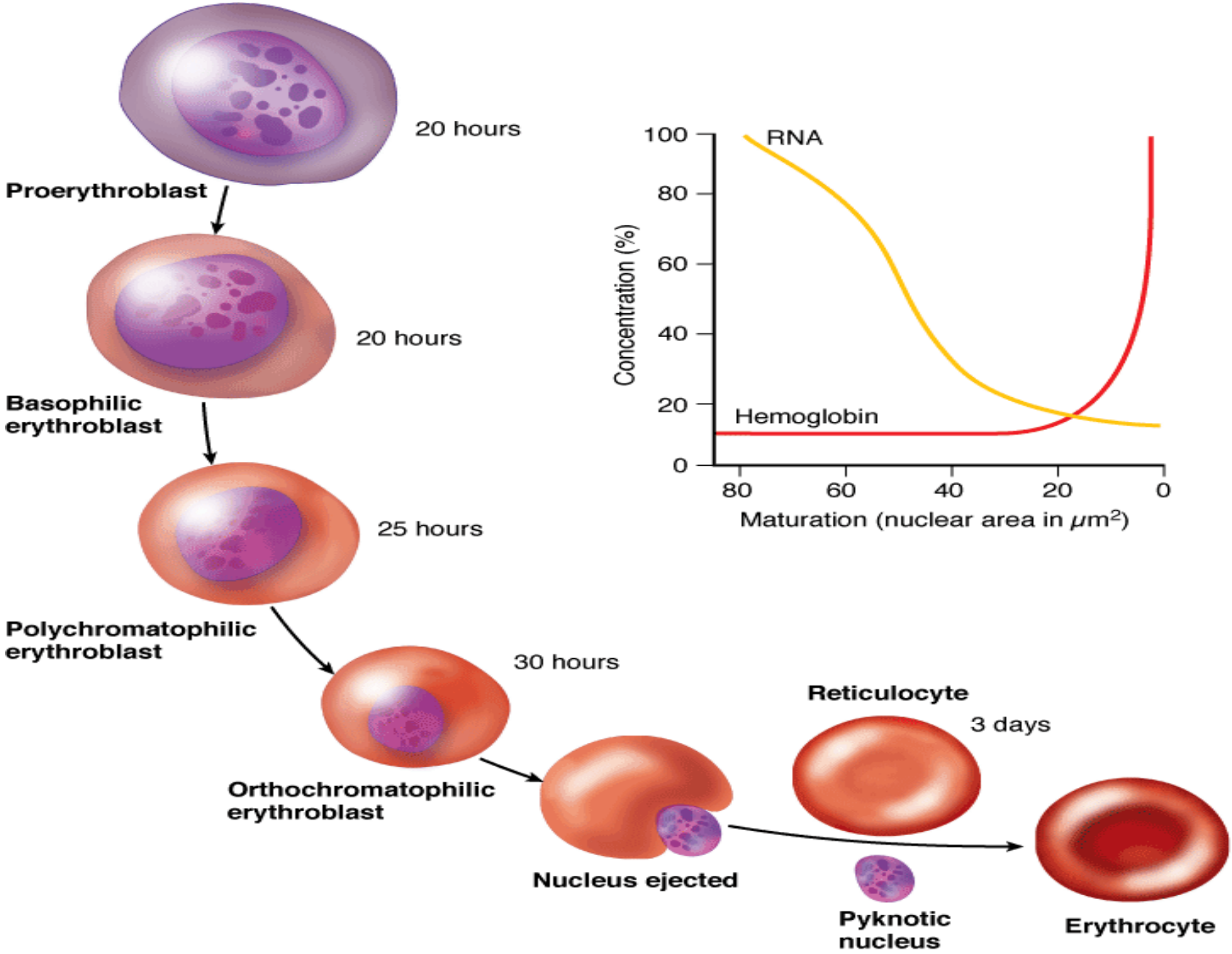




84% of the cells in the human body are red blood cells

Approximately 2.4 million new erythrocytes are produced per second in human adults

Carries oxygen from the lungs to the tissues



Source: Mescher AL: *Junqueira's Basic Histology: Text and Atlas, 12th Edition:* http://www.accessmedicine.com

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Red Blood Cells

Structure:

- Large surface area
- Contains haemoglobin which picks up oxygen
- Has no nucleus to make room for more oxygen

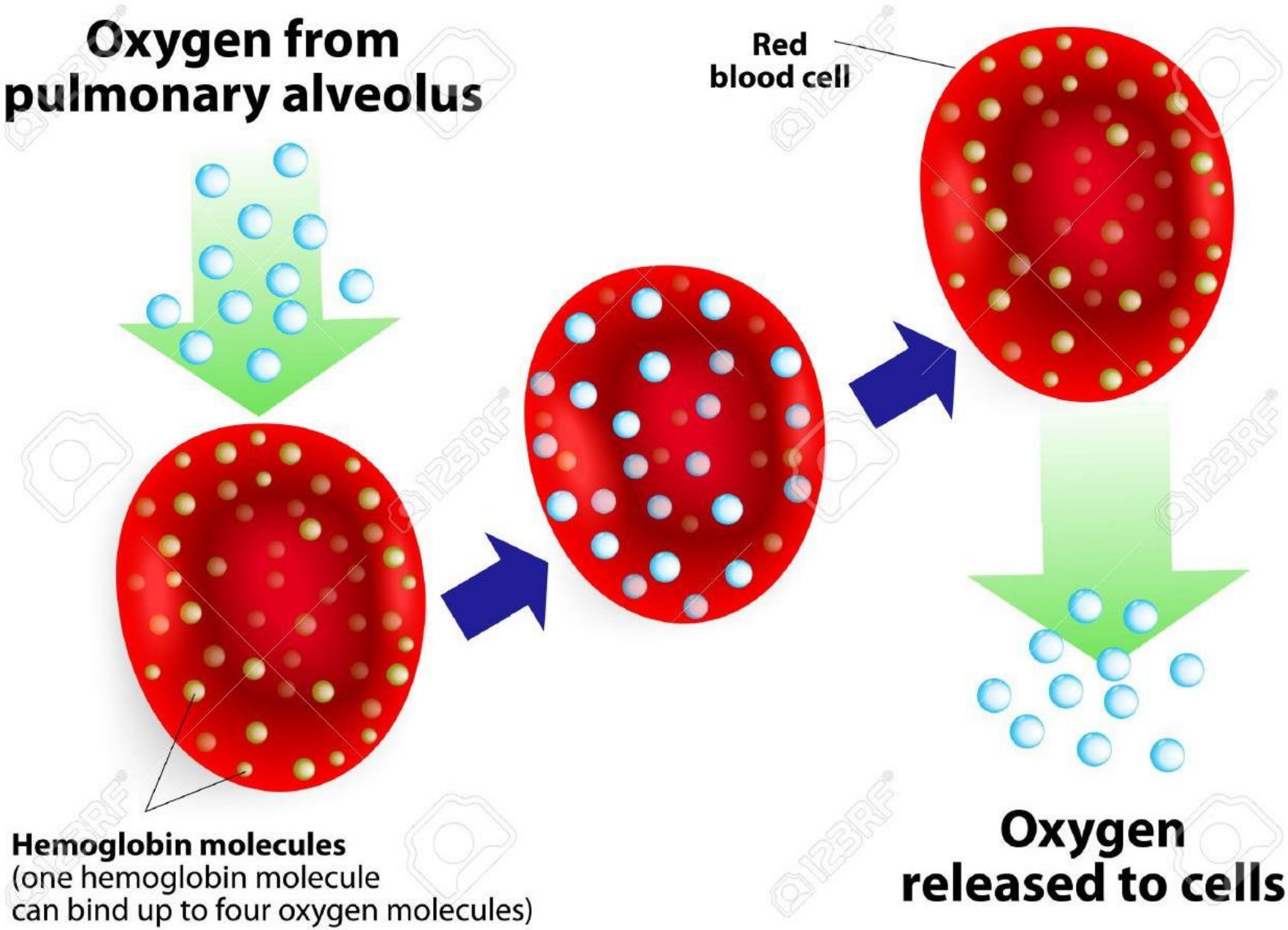


Function:

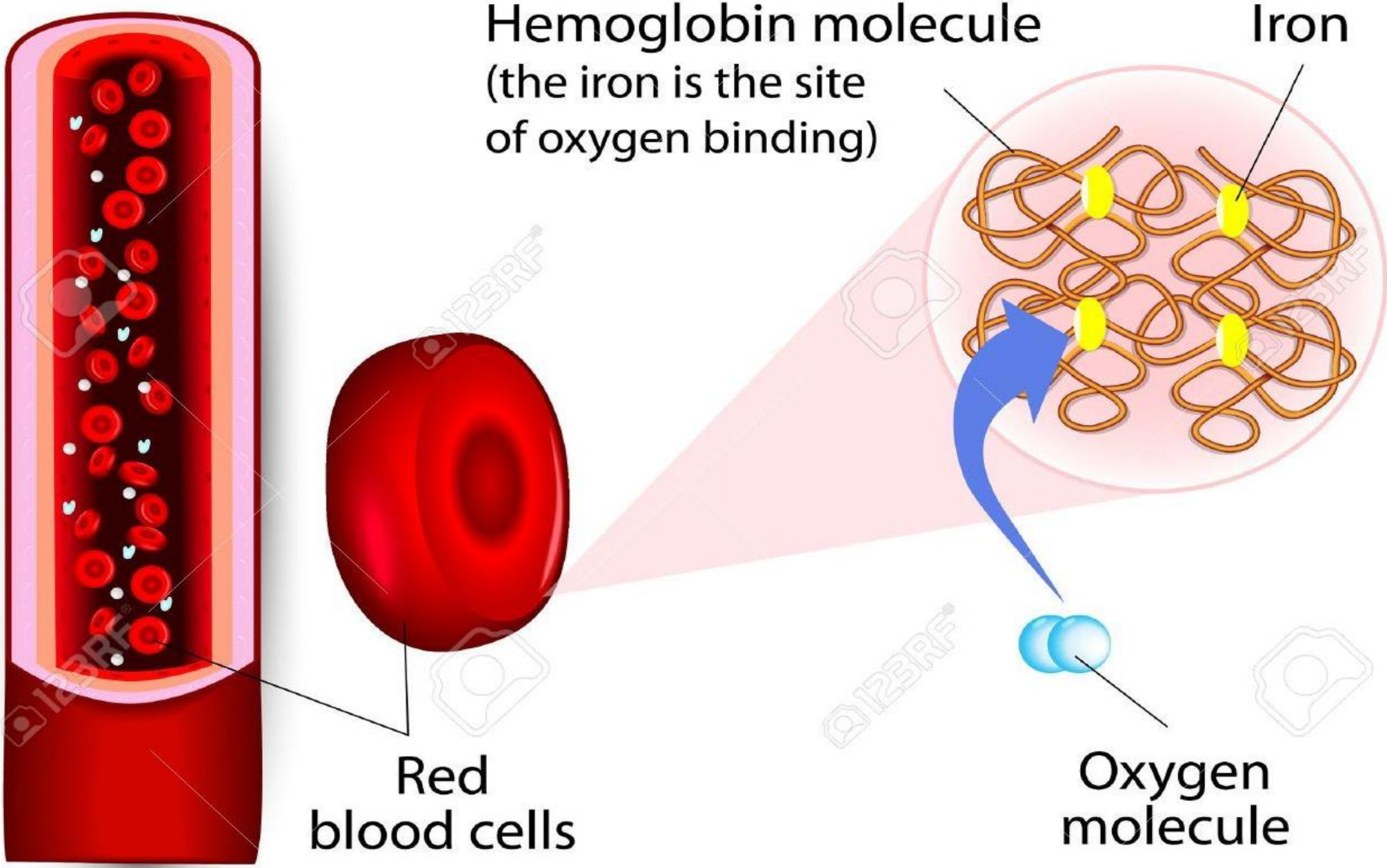
- Carries oxygen from the lungs to the body and carbon dioxide from the body back to the lungs.

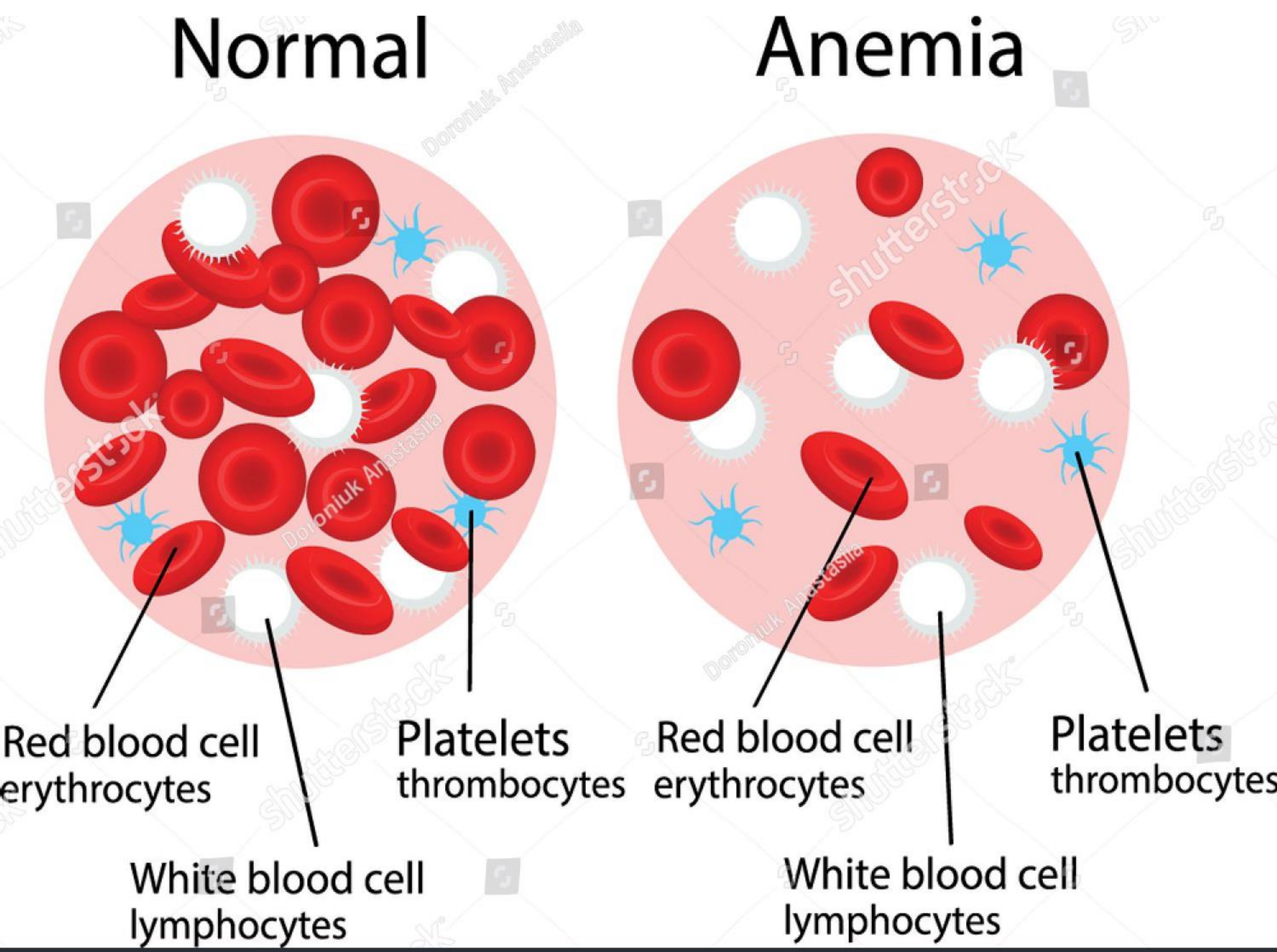
Red blood cell




Oxygen from pulmonary alveolus
Red blood cell
Hemoglobin molecules
(one hemoglobin molecule can bind up to four oxygen molecules)
Oxygen released to cells

HEMOGLOBIN


Hemoglobin molecule
(the iron is the site
of oxygen binding)
Iron
Red
blood cells
Oxygen
molecule


Normal
Anemia
Red blood cell
erythrocytes
Platelets
thrombocytes
White blood cell
lymphocytes
Red blood cell
erythrocytes
Platelets
thrombocytes
White blood cell
lymphocytes

Adult Reference Ranges for Red Blood Cells

Measurement (units)	Men	Women
Hemoglobin (gm/dL)	13.6–17.2	12.0–15.0
Hematocrit (%)	39–49	33–43
Red cell count (10^6 /μL)	4.3–5.9	3.5–5.0
Reticulocyte count (%)	0.5–1.5	
Mean cell volume (μm^3)	82–96	
Mean corpuscular hemoglobin (pg)	27–33	
Mean corpuscular hemoglobin concentration (gm/dL)	33–37	
RBC distribution width	11.5–14.5	

NORMAL HEMOGLOBIN VALUES FOR FULL-TERM AND PREMATURE INFANTS		
	FULL TERM g/dl of Blood	PREMATURE g/dl of Blood
Birth	19.3	Slightly less than Full Term
0.5 Months	16.6	15.4
1 month	13.9	11.6
Age at Hemoglobin Nadir	9-12 weeks	6-10 weeks
Mean Hemoglobin at Nadir	11.2	9.4
4 Months	12.2	11.7
6 Months	12.5	12.4

Age	Normal level of hemoglobin (gram per deciliter of blood)
Newborns	17 - 22 g/dl
1 week old	15 - 20g/dl
1 month old	11 - 15 g/dl
6 months – 2 years	11 - 13 g/dl
2 - 6 years	11.5 – 13.5 g/dl
6 – 12 years	12.5 – 14.5 g/dl
12 – 18 years boys	14.5 g/dl
12- 18 years girls	14.0 g/dl

?What is Anemia

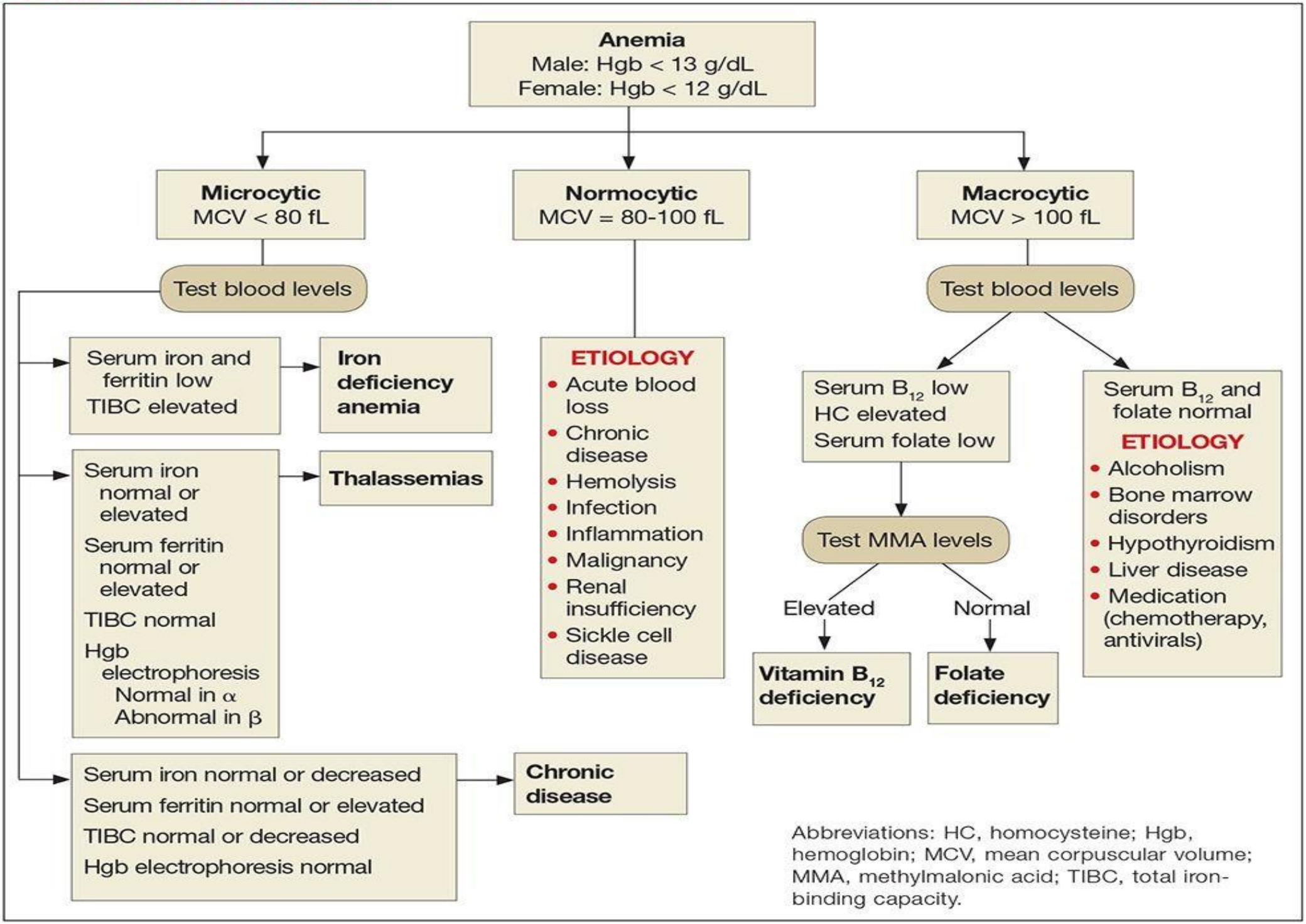
- A condition characterized by a decrease in RBCs mass, hemoglobin or RBCs count.

- Adult male: Hb < 13 g/dL, or Hct.: 41%.
- Adult Females: Hb < 12 g/dL, or Hct.: <36%.

- Normal:
  - Adult male: Hb: 13.5 – 17.5 g/dl, Hct.: 40% - 54%
  - Adult Females: Hb: 12-15 g/dL, Hct.: 37% - 47%

FIGURE

Classification of Anemia



Abbreviations: HC, homocysteine; Hgb, hemoglobin; MCV, mean corpuscular volume; MMA, methylmalonic acid; TIBC, total iron-binding capacity.

:Clinical Features

- History:
  - Symptoms of anemia:
    - Fatigue, malaise, dyspnea, decreasrd exercises tolerance, palpitation, headache, dizziness, syncope…
  - History of acute or chronic bleeding.
  - Diet.
  - Family history.

:Clinical Features

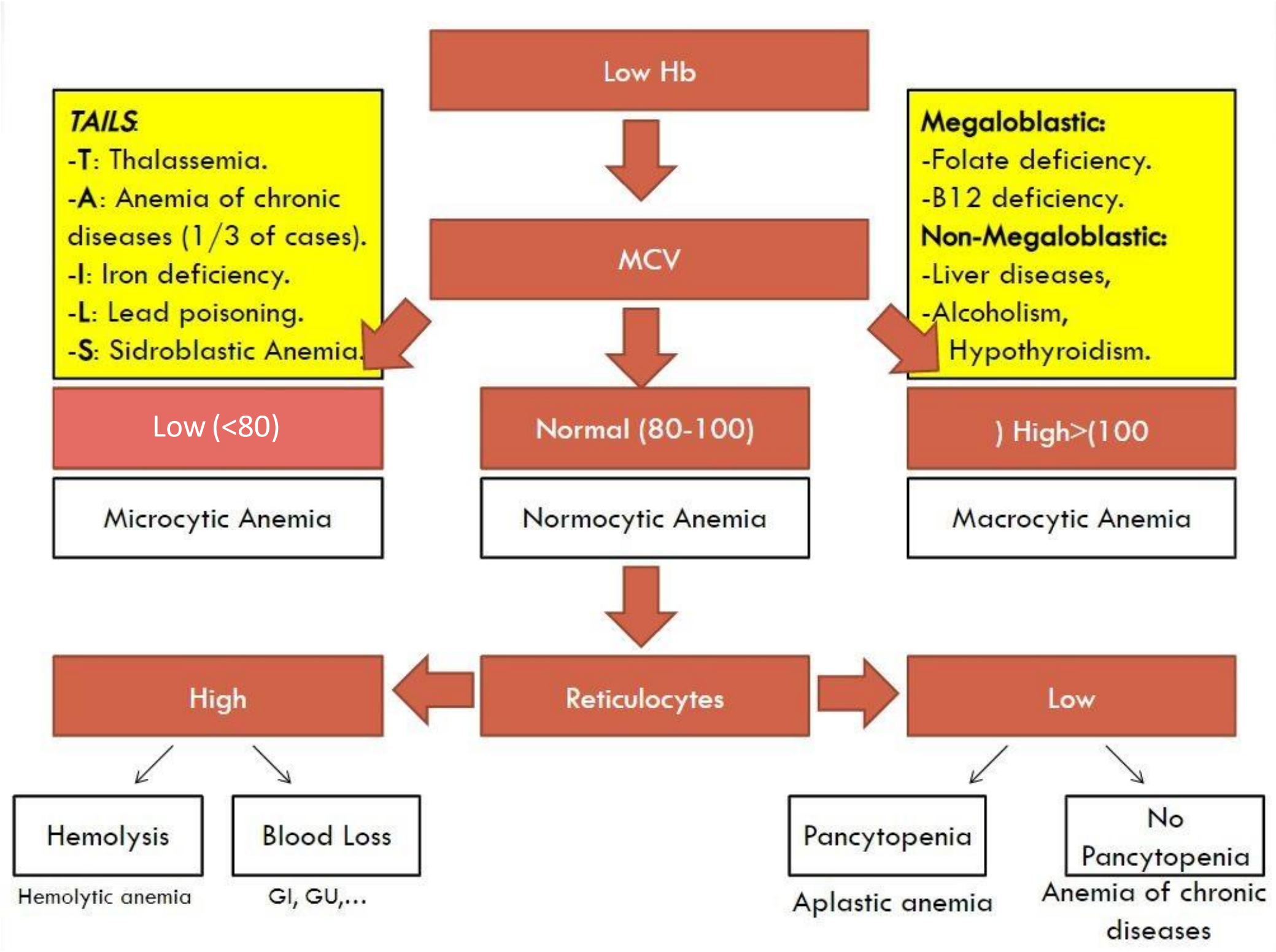
- History:
  - Rule out pancytopenia:
    - Recurrent infection, mucosal bleeding, bruising,..
  - Drug history.

:Clinical Features

- Physical Examination:
  - Vital signs:
    - Pulse?? Collapsing pulse.
    - Orthostatic hypotension.
  - Pallor: palmar creases, conjunctiva, mucus membranes,..
  - Jaundice: hemolysis.
  - CVS: slow systolic flow murmur, signs of CHF.

:Lab. Investigations

- CBC: with differential count, Hb, MCV, RDW
- Reticulocyte count.
- Blood film.


Low Hb
MCV
TAILS:
-T: Thalassemia.
-A: Anemia of chronic diseases (1/3 of cases).
-I: Iron deficiency.
-L: Lead poisoning.
-S: Sidroblastic Anemia.
Megaloblastic:
-Folate deficiency.
-B12 deficiency.
Non-Megaloblastic:
-Liver diseases,
-Alcoholism,
Hypothyroidism.
Low (<80)
Normal (80-100)
) High>(100
Microcytic Anemia
Normocytic Anemia
Macrocytic Anemia
Reticulocytes
High
Low
Hemolysis
Blood Loss
Pancytopenia
No Pancytopenia
Hemolytic anemia
GI, GU,...
Aplastic anemia
Anemia of chronic diseases

MICROCYTIC ANEMIA

***TAILS*:**

-**T**: Thalassemia.

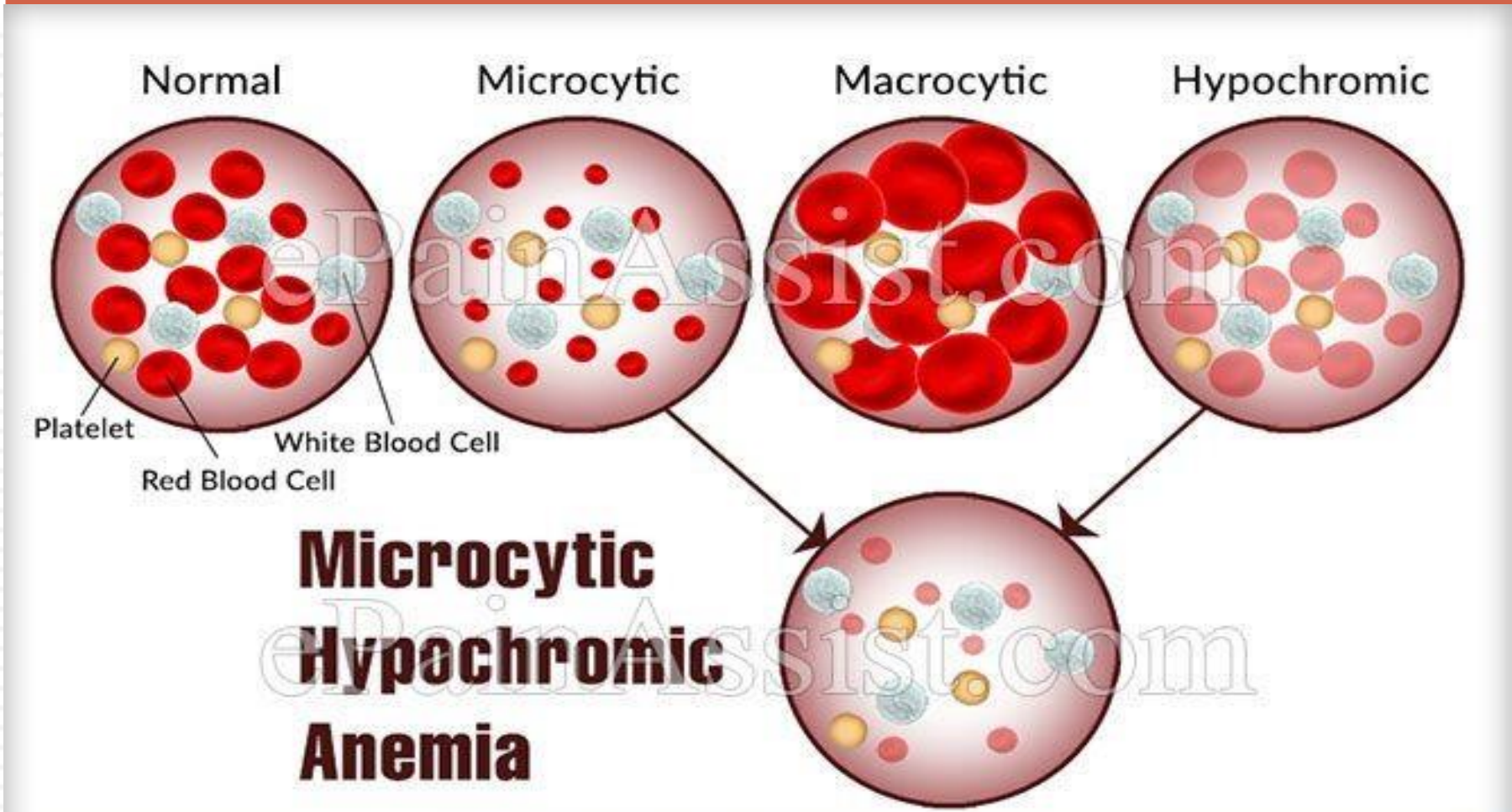
-**A**: Anemia of chronic diseases (1/3 of cases).

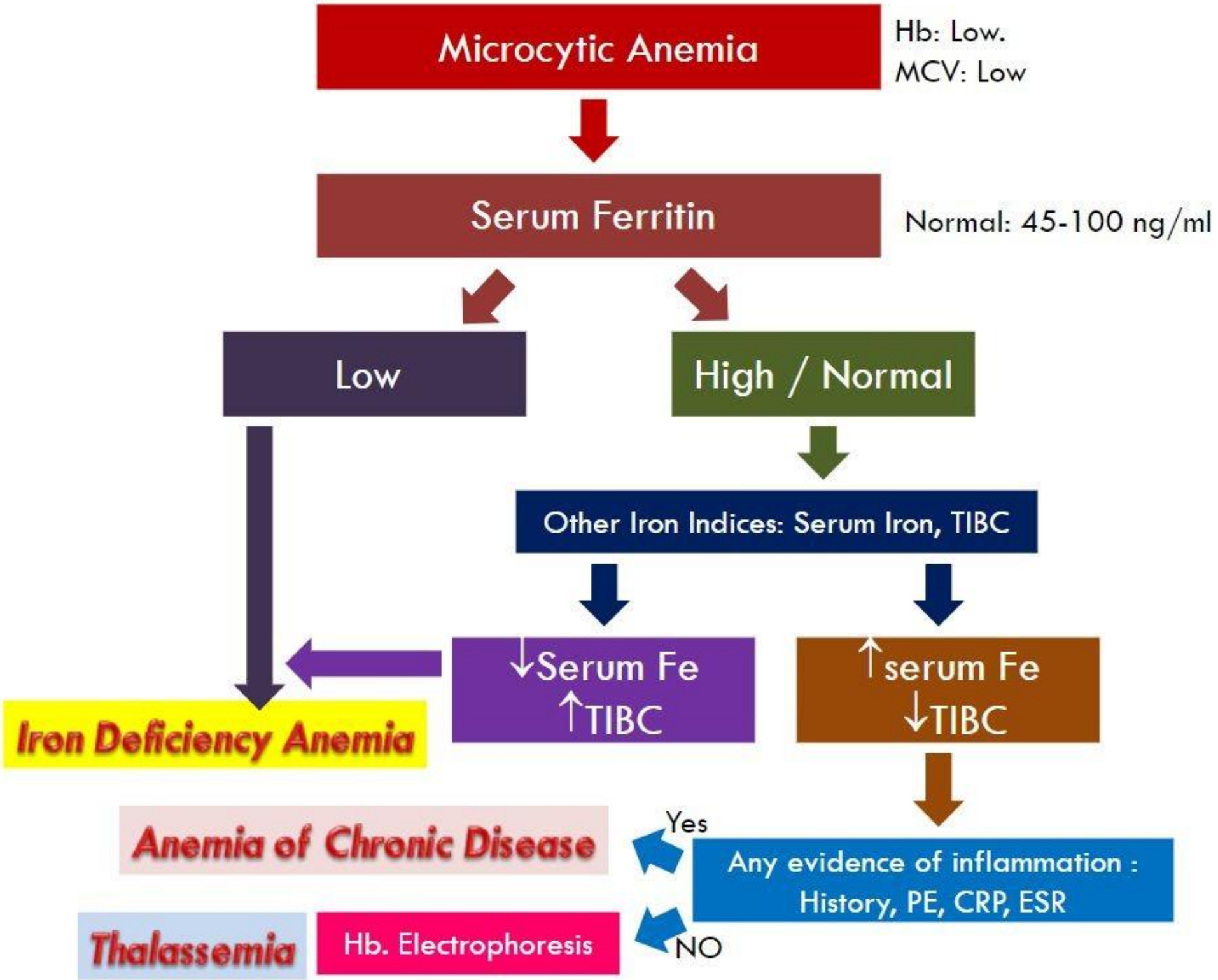
-**I**: Iron deficiency.

-**L**: Lead poisoning.

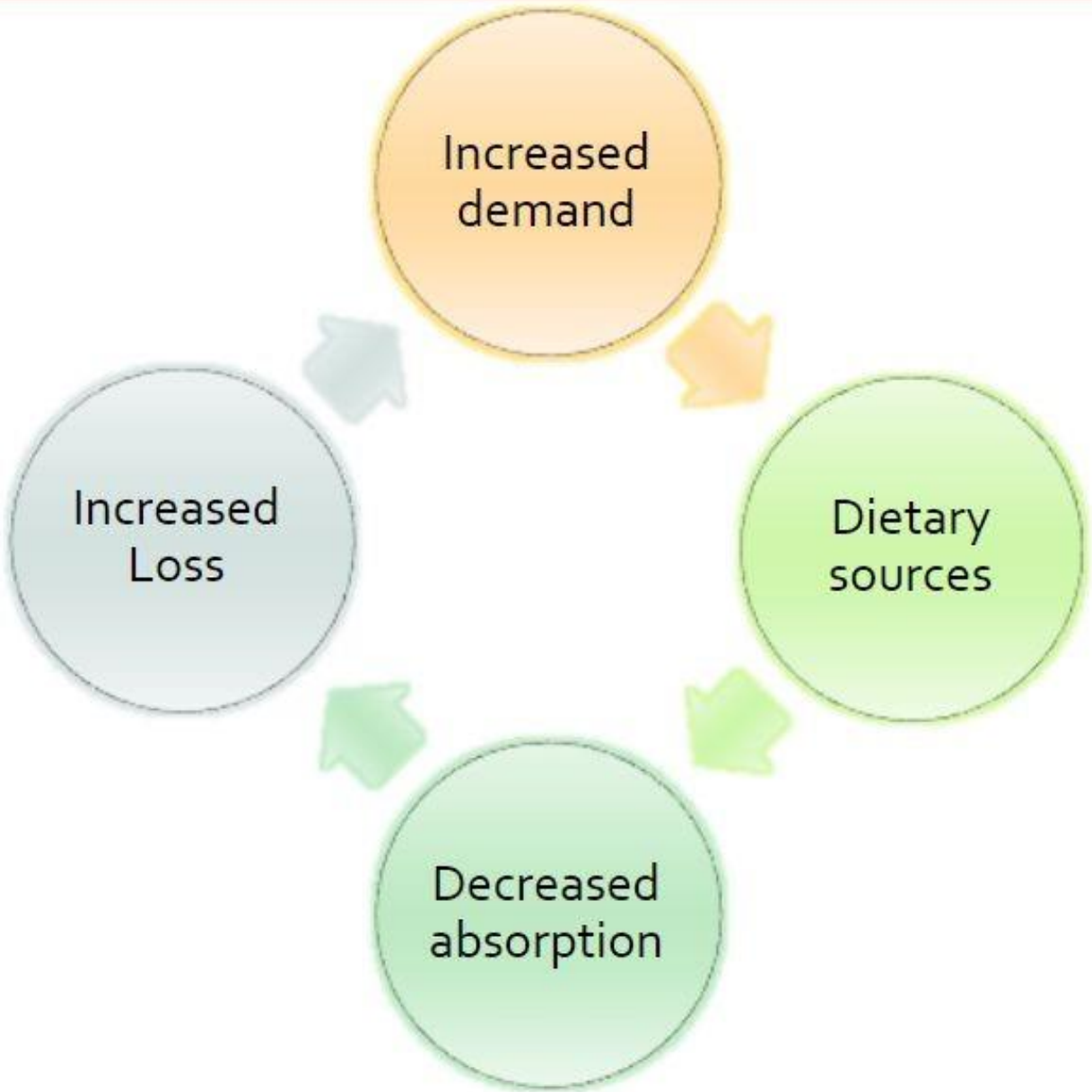
-**S**: Sidroblastic Anemia.

Approach to Microcytic Anemia




Microcytic Anemia
Hb: Low.
MCV: Low
Serum Ferritin
Normal: 45-100 ng/ml
Low
High / Normal
Other Iron Indices: Serum Iron, TIBC
↓Serum Fe
↑TIBC
↑serum Fe
↓TIBC
Iron Deficiency Anemia
Anemia of Chronic Disease
Yes
Any evidence of inflammation :
History, PE, CRP, ESR
Thalassemia
Hb. Electrophoresis
NO

:IRON DEFICIENCY ANEMIA



:IRON DEFICIENCY ANEMIA

- CAUSES:
  - ***Increased demand:***
    - E.g., pregnancy, growing child,
  - ***Dietary deficiency:***
    - Cow's milk: infant.
    - Low meat.
  - ***Absorption imbalance:***
    - Post-gasterectomy.
    - Malabsorption (IBD, celiac disease, atrophic gasteritis).
    - Factors decreasing iron absorption e.g., tea,…

:IRON DEFICIENCY ANEMIA

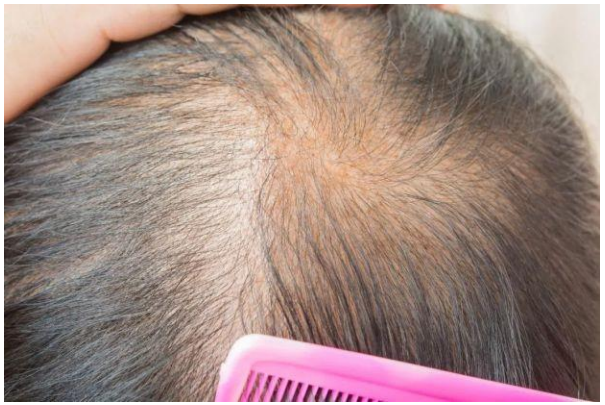
- CAUSES:
  - ***Increased Iron Loss:***
    - Hemorrhage:
      - Peptic Ulcer,
      - GI cancer,
      - GI parasites,
      - Menorrhagia,

:IRON DEFICIENCY ANEMIA

- Clinical Features:
  - Symptoms due to anemia:
    - Fatigue, weakness, irritability, exercise intolerance, syncope, dyspnea, headache, palpitations, postural dizziness, tinnitus, feeling cold, confusion / loss of concentration..
  - Pallor…

:IRON DEFICIENCY ANEMIA

- Clinical Features:
  - Brittle hair.



Koilonychia

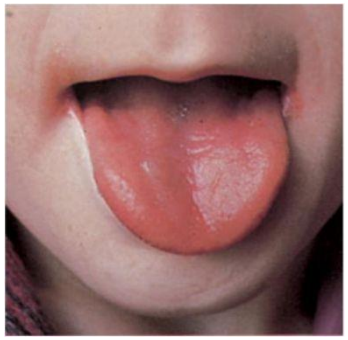


- Dysphagia:

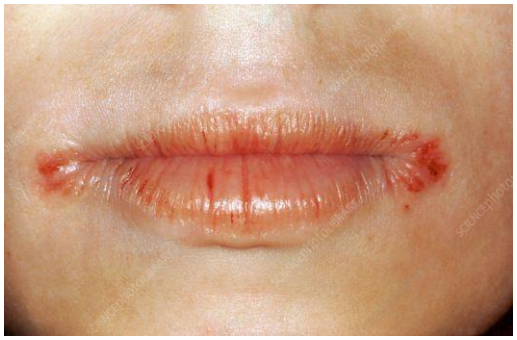
Plummar-Vinson's syndrome



- Glossitis:



- Angular Stomatitis:



:IRON DEFICIENCY ANEMIA

- Clinical Features:
  - Pica:
    - Appetite for non-food substances.

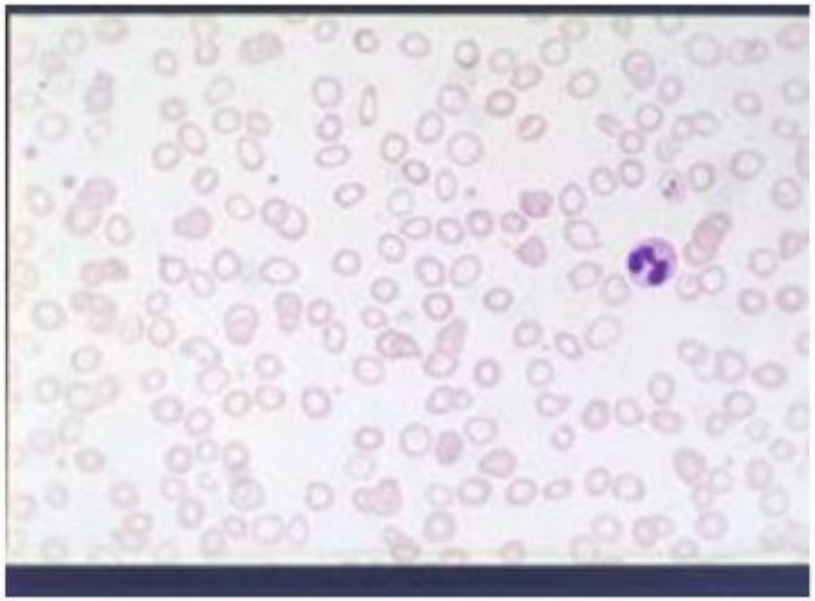






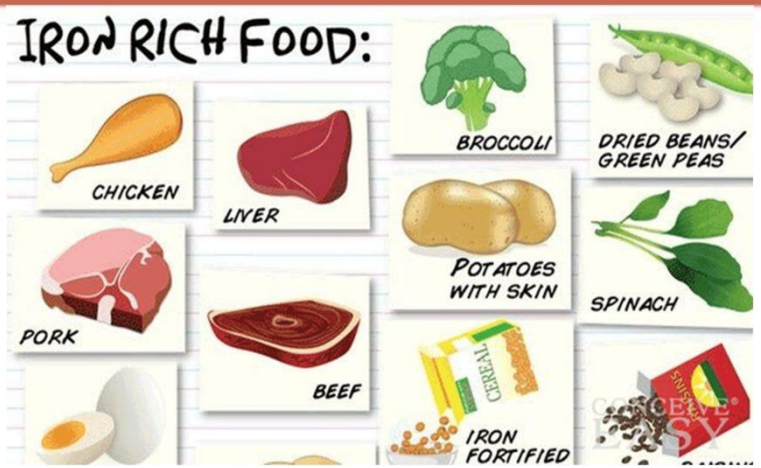
:IRON DEFICIENCY ANEMIA

- Investigations:
  - Iron Indices:
    - Low ferritin (<45µg/dL).
    - ↓serum iron, ↑TIBC.
  - Peripheral blood film:
    - Microcytic, hypochromic.
    - Anisocytosis, pencil forms,
    - Target cells.



:IRON DEFICIENCY ANEMIA

- Management:
  - ***Treat underlying cause.***
  - ***Iron supplements:***
    - Oral: ferrous sulfate 325mg tid.
    - IV iron: if patient cannot tolerate oral iron.
  - ***Monitoring response:***
    - Reticulocyte will begin to increase within 1 week.
    - Hb normalizes by 10g/dL per week.
    - Fe supplements are required for 4-6 months to replenish iron stores.



:Prevention of Iron Deficiency Anemia

- Breastfeeding should be encouraged with the addition of supplemental iron at 4 mo. of age (12mg daily).
- Infants who are not breastfed should only receive iron-fortified formula (12 mg of iron per liter) for the first year, and thereafter cow's milk should be limited to <20-24 oz daily.

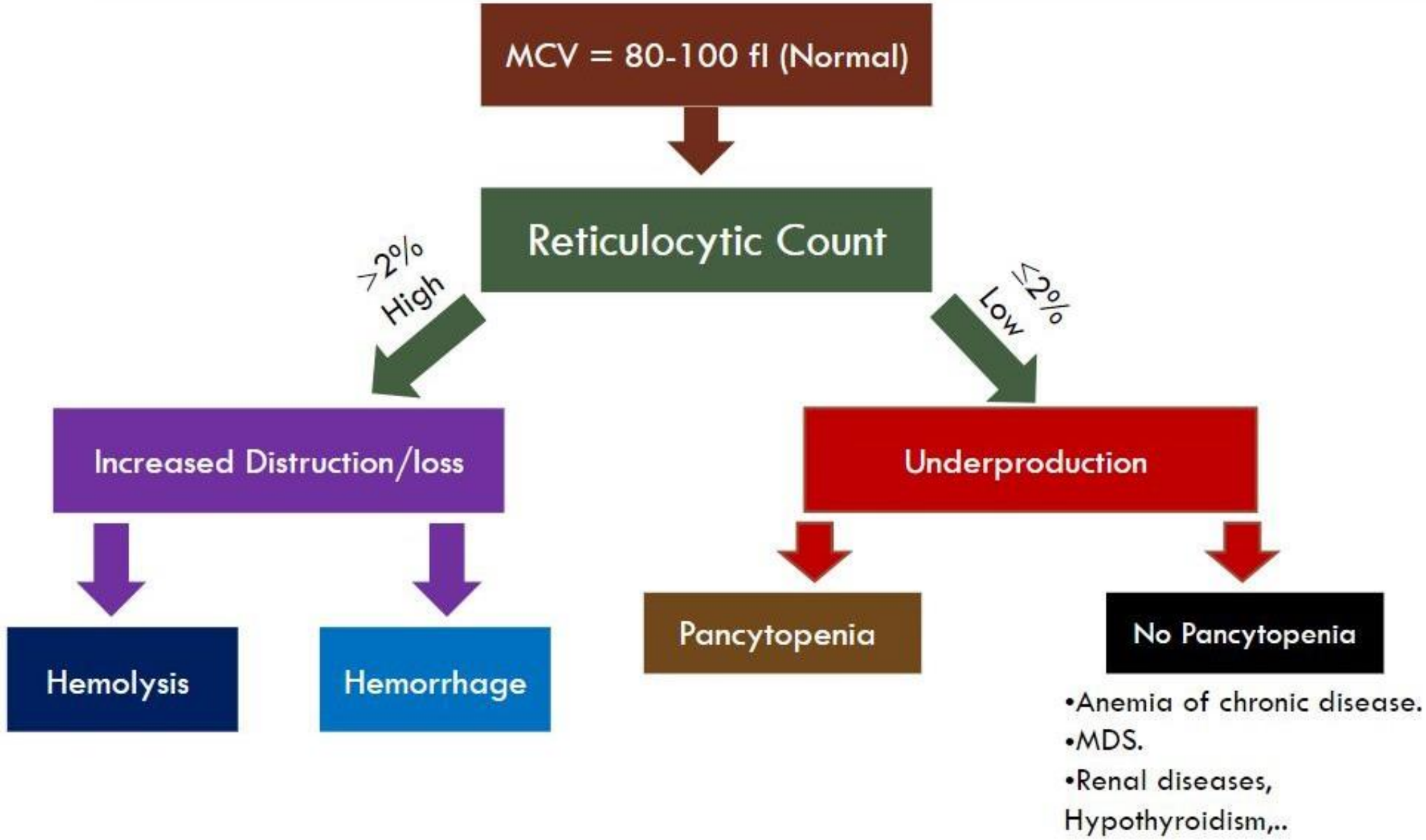


:Prevention of Iron Deficiency Anemia

During the first 2 years of life to ensure healthy growth and development, **optimal infant and young child feeding practices include**:

- Exclusive breastfeeding* up to 6 months of age
- Introduction of foods to complement the nutrients in breast milk at 6 months of age
- Continuation of breastfeeding for 2 years.

:NORMOCYTIC ANEMIA



:NORMOCYTIC ANEMIA

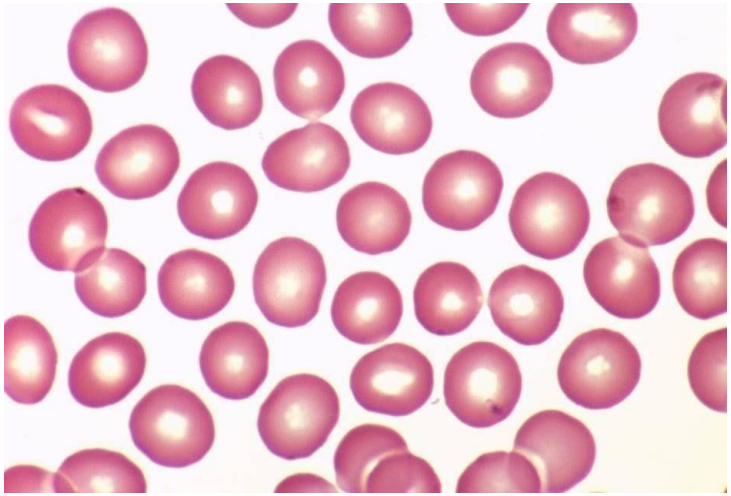
- CAUSES:

Increased Loss

- Acute Hemorrhage.
- Hemolysis.

Decreased Production

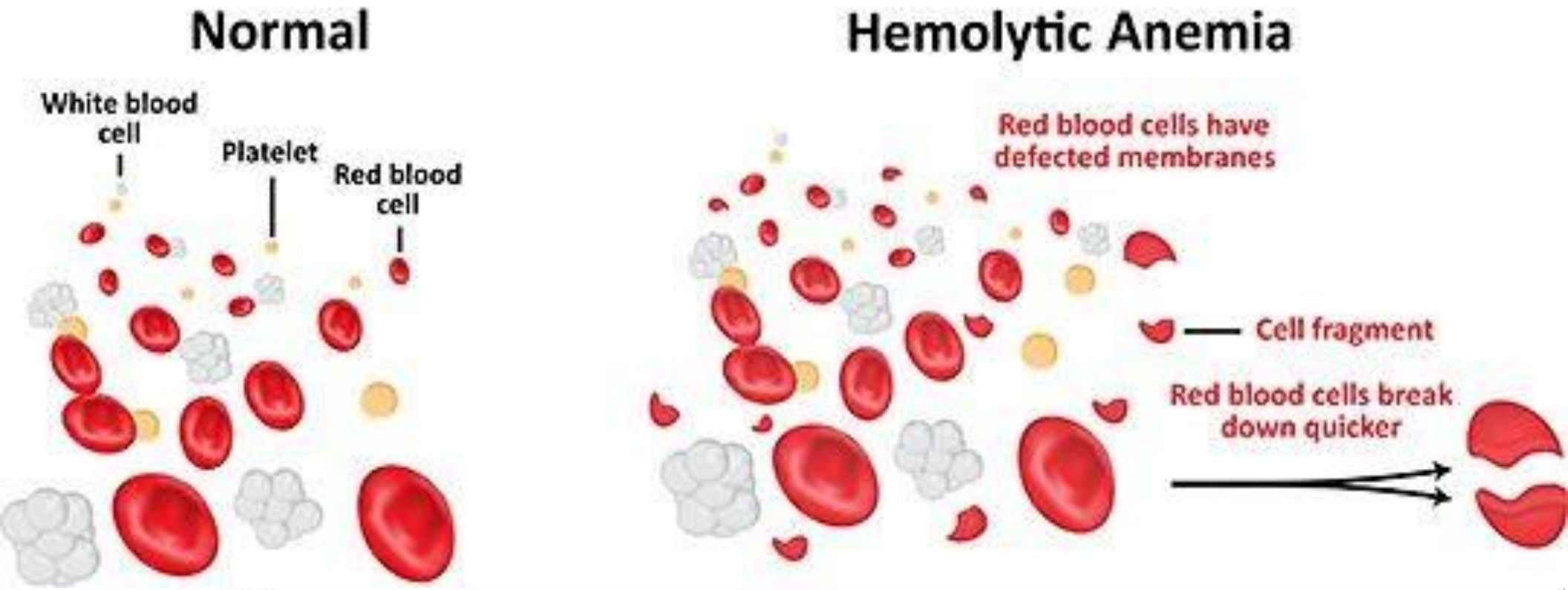
- Bone Marrow Failure.
- Aplstic anemia.
- Chronic Diseases.



:NORMOCYTIC ANEMIA

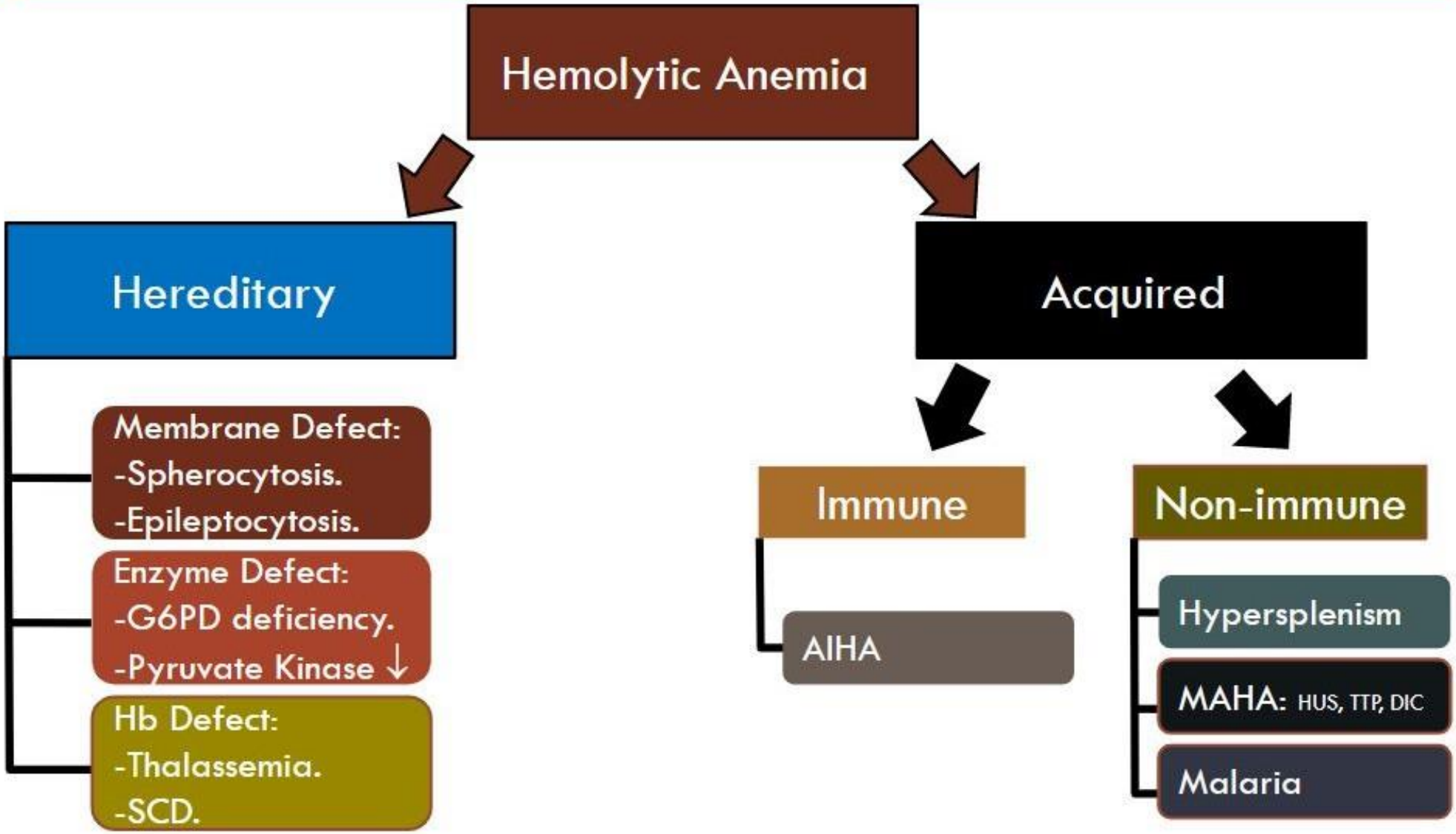
- CAUSES:
  - **A B C D** :
    - **A**: Acute blood loss.
    - **B**: Bone Marrow Failure.
    - **C**: Chronic Disease.
    - **D**: Distruction (hemolysis).

Hemolytic Anemia



Hemolytic Anemia

:HEMOLYTIC ANEMIA - Classification



:HEMOLYTIC ANEMIA - Classification

- ***INTRA-VASCULAR HEMOLYSIS:***
  - results from rupture or lysis of red blood cells **within** the circulation.
  - E.g., : G6PD, TTP, DIC,…

- ***EXTRA-VASCULAR HEMOLYSIS:***
  - Results from phagocytosis of abnormal RBCs in spleen, liver and bone marrow.
  - E.g., AIHA, spherocytosis,..

:HEMOLYTIC ANEMIA

- CLINICAL FEATURES:
  - Jaundice.
  - Dark urine.
  - Cholelithiasis.
  - Iron overload: extravascular hemolysis.
  - Iron deficiency: intravascular hemolysis.
  - Aplastic crisis: infection with Parvo B19.







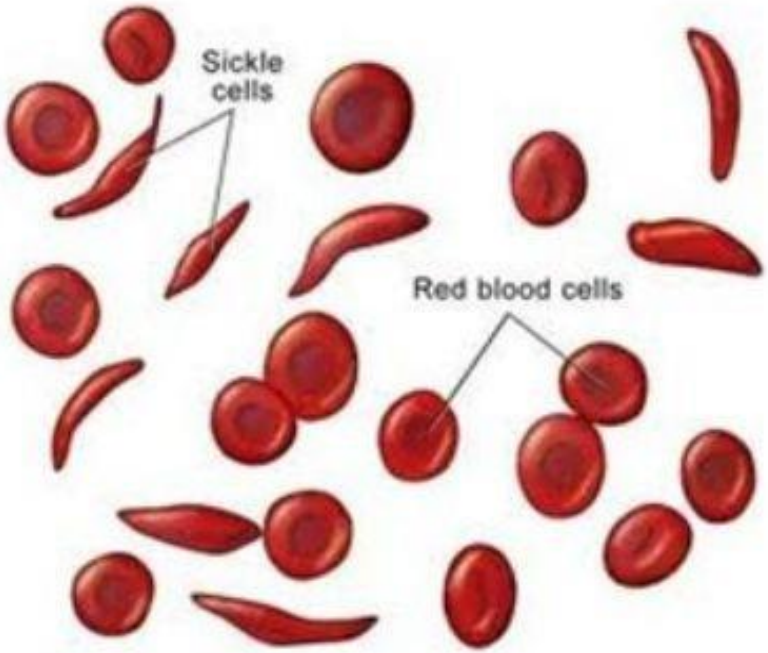
:HEMOLYTIC ANEMIA

- INVESTIGATIONS:
  - ↑ Reticulocyte count.
  - ↓ Haptoglobin.
  - ↑ Un-conjugated bilirubin.
  - ↑ Urobilinogen.
  - ↑ LDH.
  - Blood film.
  - Coomb's test: immune mediated hemolysis.

Hemolytic Anemia

Sickle Cell Anemia

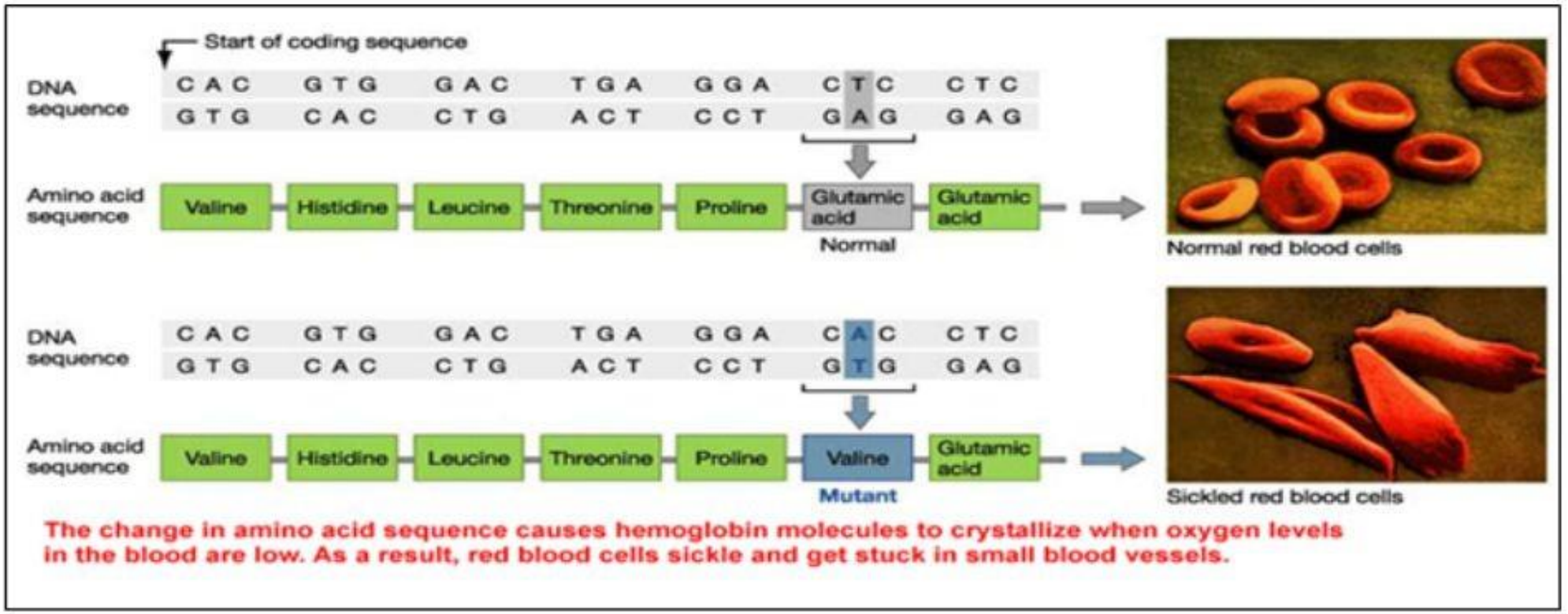
SCD



:Sickle Cell Anemia

- Autosomal Recessive.

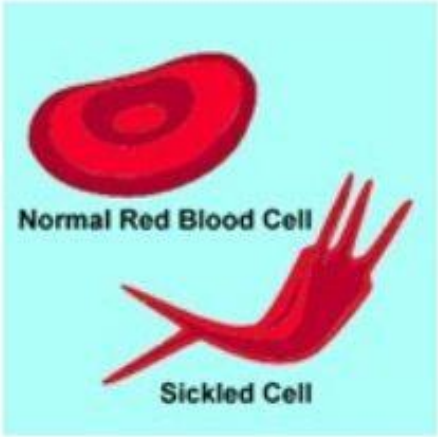
- Mutation in the 6th amino acid in β-globin chain in which glutamic acid is replaced by Valine--- HbS.

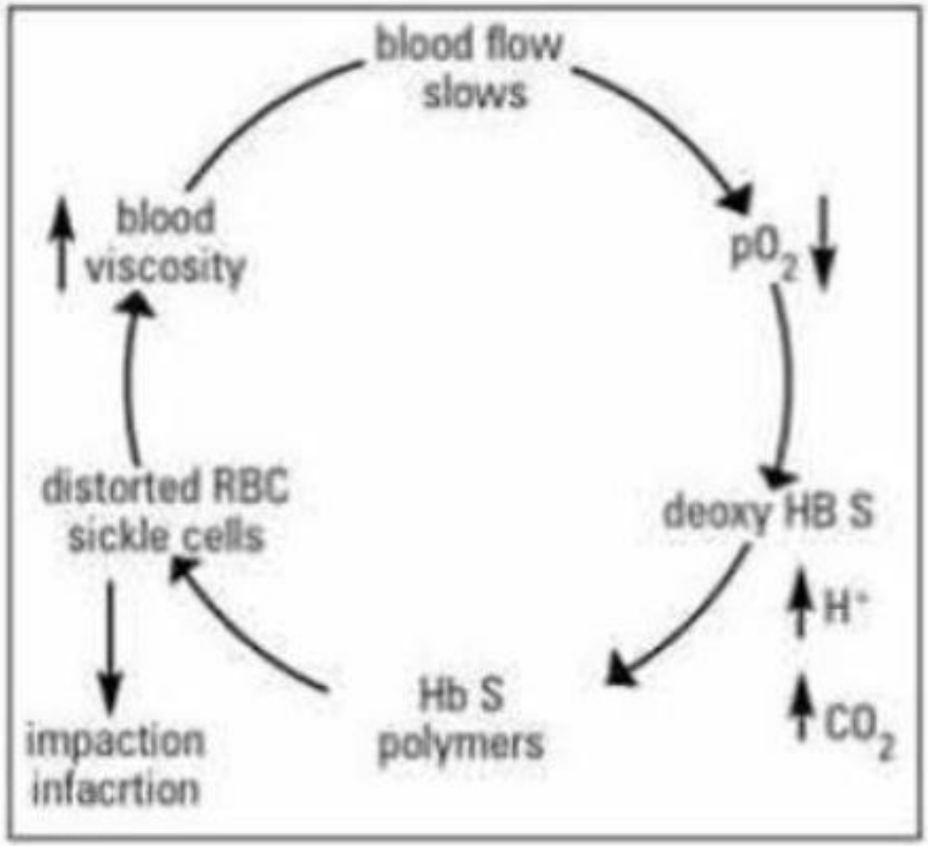


:Sickle Cell Anemia

- Pathophysiology:

- at low pO2, deoxy HbS polymerizes, leading to rigid crystal-like rods that distort membranes ..... 'sickles'





:Sickle Cell Anemia

- Clinical Features:
  - HbAs: appears normal.
  - HbSS (homozygous):
    - Chronic hemolytic anemia.
    - Jaundice.
    - Growth retardation in child (skeletal changes).
    - Spleenomegally in children… (but atrophy in adults).
    - Crises.

:Sickle Cell Anemia

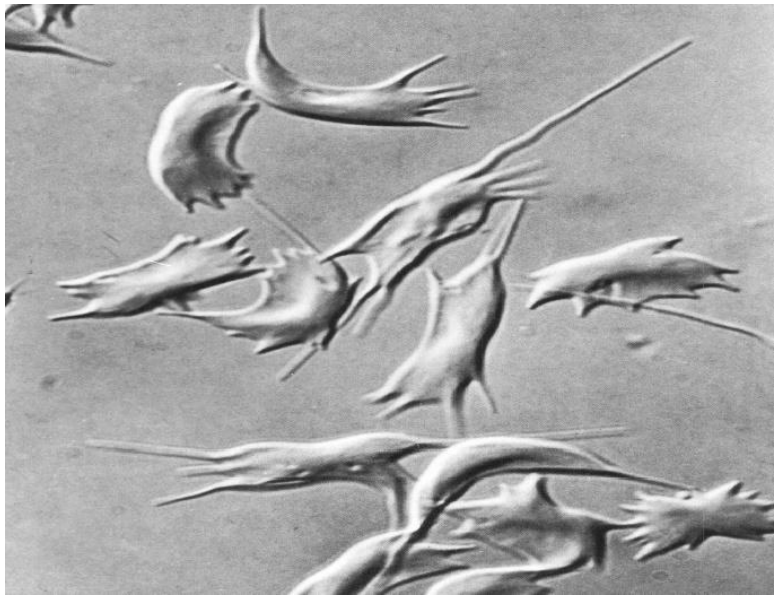
- Sickle Cell Crises:
  - ***Vaso-occlusive Crises:***
    - Pain: back, abdomen, extremities,…
    - Acute chest syndrome: pneumonia like.
  - ***Priapism.***
  - ***Aplastic Crises:***
    - Precipitated by toxins or infections (B19 virus).
  - ***Splenic Sequestration Crises:***
    - In children.
    - Significant pooling of blood to spleen resulting in ↓Hb and shock.
    - Rare in adults: already have functional asplenism…

:Sickle Cell Anemia

- ***Functional Asplenism:***
  - Increases susceptibility to infection by encapsulated organisms:
    - *Strept. pneumoniae.*
    - *N. meningitidis.*
    - *H. influenzae.*
    - *Salmonella* (osteomyelitis).
  - Child with SCD should be receive pneumococcal, Hib, Meningococcal vaccines.

:Sickle Cell Anemia

- Triggers of Crises:
  - Hypoxia,
  - Acidosis.
  - Infection.
  - Fever.
  - Dehydration.



:Sickle Cell Anemia

- Management:
  - Hydroxyurea:
    - To ↑ HbF which has a higher affinity to O2.
  - Folate:
    - to prevent folate deficiency.

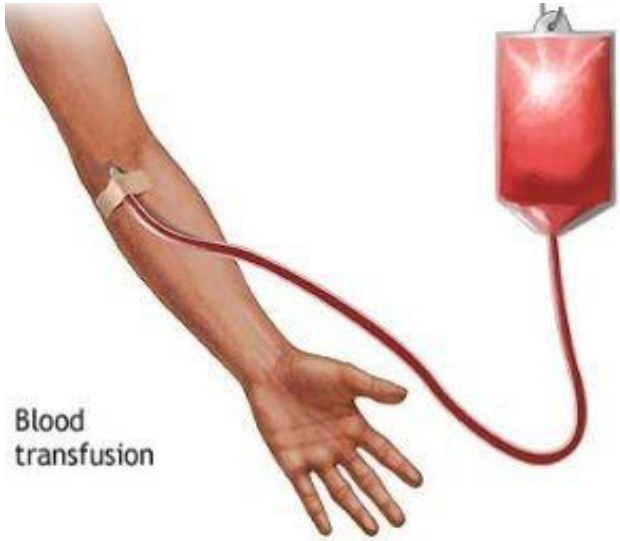
:Sickle Cell Anemia

- Management:
  - Management of Vaso-occlusive Crises:
    - Oxygen.
    - Hydration.
    - Antimicrobial.
    - Analgesic.
    - Mg.
    - Transfusion if indicated.



:Sickle Cell Anemia

- Management:
  - Management of Vaso-occlusive Crises:
    - Indications of Transfusion:
      - Acute Chest Syndrome.
      - Stroke.
      - BM necrosis.
      - Priapism.
      - CNS crises.

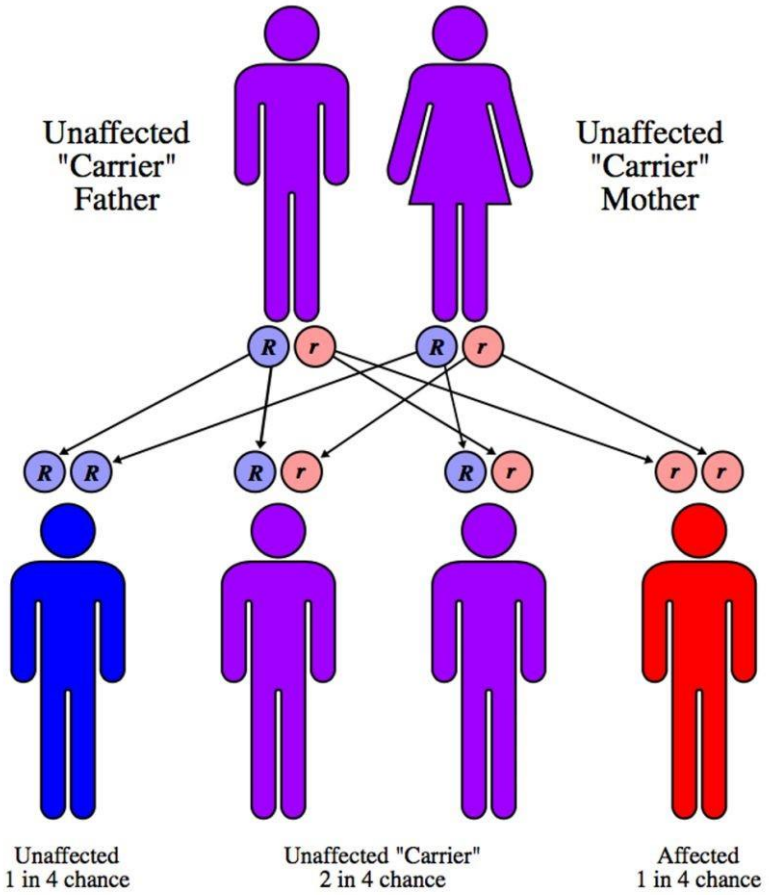

Blood transfusion

:Sickle Cell Anemia

- Management:
  - Avoid conditions that induce crises.
  - Vaccination.
  - Prophylactice penicillin (3 months – 5 years age).

Hemolytic Anemia

Thalassemia



:Thalassemia

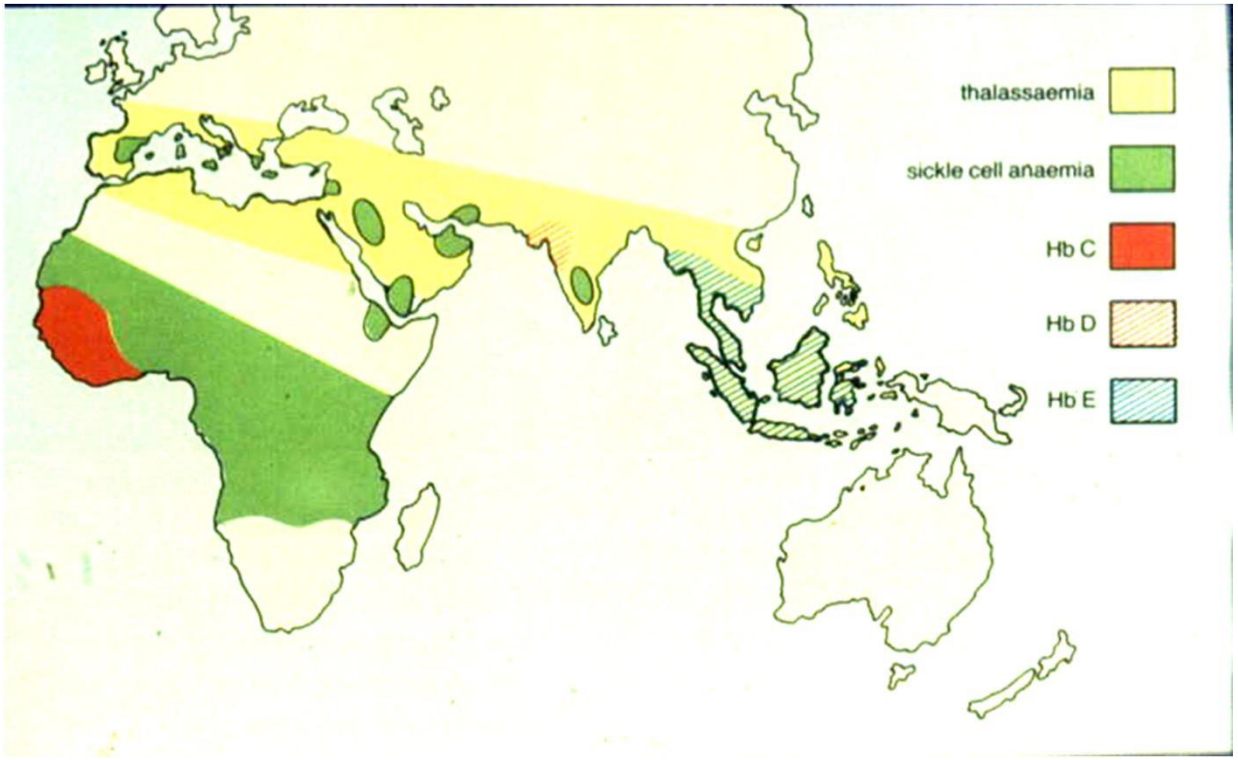
- Disorders involving ↓ or absent production of normal globin chains of hemoglobin.

- **α-thalassemia** is caused by a mutation of one or more of the **four** genes for α-hemoglobin;

- **β-thalassemia** results from a mutation of one or both of the **two** genes for β-hemoglobin.

:Thalassemia

- Thalassemia is most common among people of **African, Middle Eastern, and Asian descent.**



α Thalassemia

- **Predominant cause of alpha thalassemias is large number of gene deletions in the α-globin genes.**
- **There are four clinical syndromes present in alpha thalassemia:**
  - **Silent Carrier State**
  - **Alpha Thalassemia Trait (Alpha Thalassemia Minor)**
  - **Hemoglobin H Disease**
  - **Bart's Hydrops Fetalis Syndrome**



:Thalassemia

- Types:

Hydrops fetalis	0/4 α	Patients die in utero.
Hemoglobin H disease	1/4 α	Patients have severe hypochromic, microcytic anemia with chronic hemolysis, splenomegaly, jaundice, and cholelithiasis. The reticulocyte count elevates to compensate, and one-third of patients have skeletal changes due to expanded erythropoiesis.

:Thalassemia

- Types:

α-thalassemia trait	2/4 α	Patients have low MCV but are usually asymptomatic.
Silent carrier	3/4 α	Patients have no signs or symptoms of disease.

Types of βThalassemia

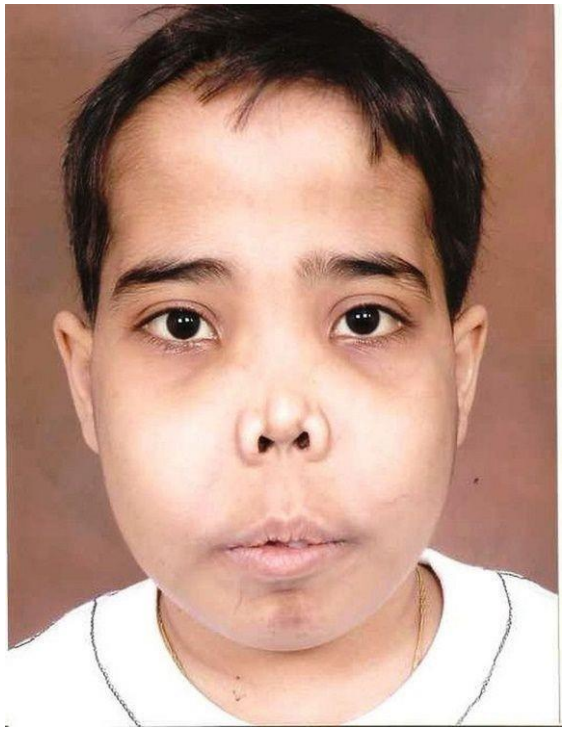
Three common types of Thalassemia:

- Thalassemia: The production of chain is mildly reduced.
- Thalassemia: The production of chain is more reduced than But NOT ABSENT. and are caused by mutation in Promoter region, 5`UTR, Cap site, Consensus sites, within Introns, 3`UTR, or Poly A site, and change in coding region.
- Thalassemia: ABSENCE of chain production. It is caused by mutation in Initiation codon, Splicing at junctions, Frameshift, Nonsense mutation.

β Thalassemia Major

- Have protrusion upper teeth and Mongoloid facial features.
- Physical growth and development delayed.





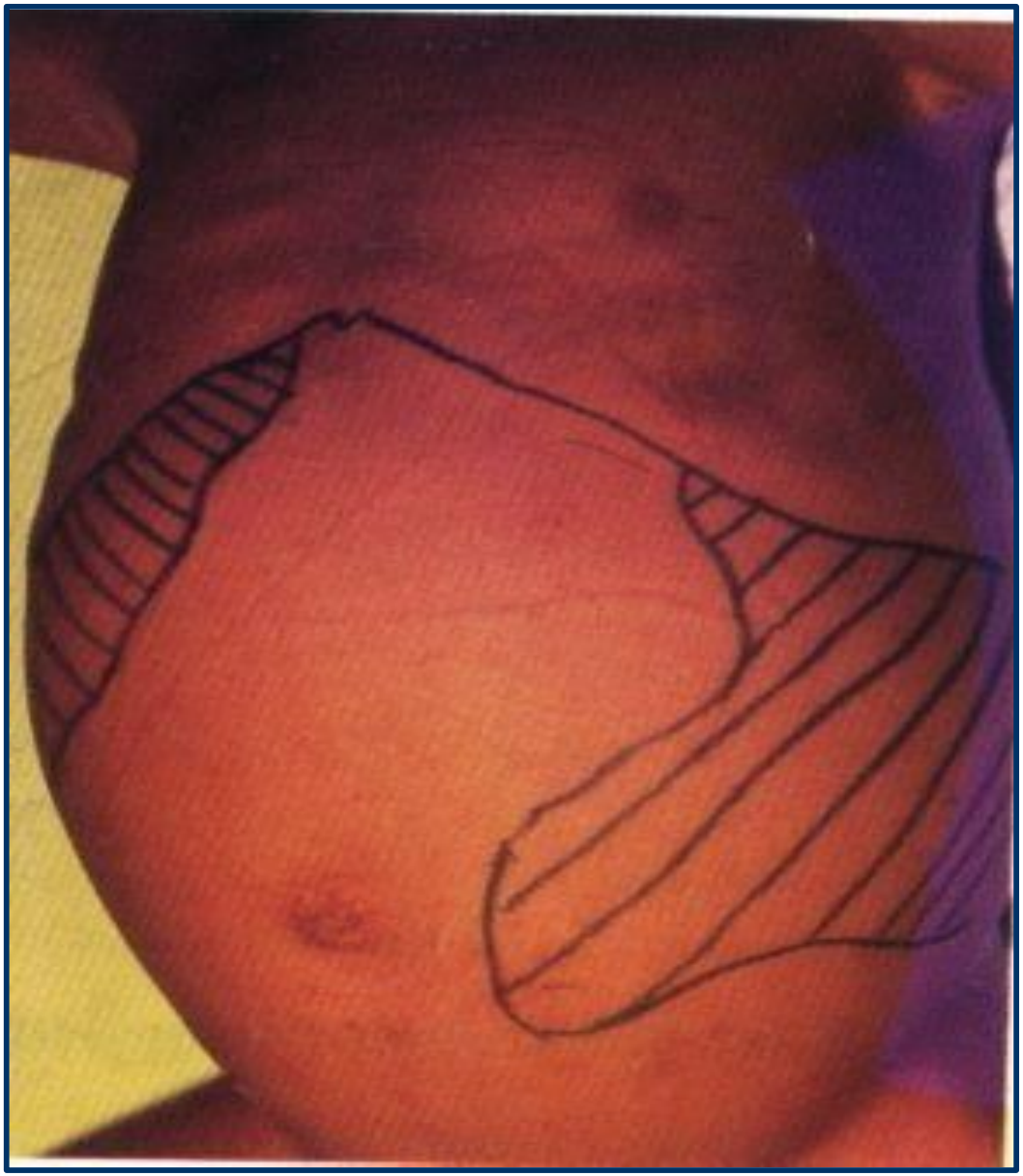
Clinical Symptoms

- Anemia, Jaundice.
- Hepatosplenomegaly.
- Bone change-->mongoloid face.
- Iron overload-->growth retardation, heart failure, DM, dark-colored skin , etc.

Thalassemia face



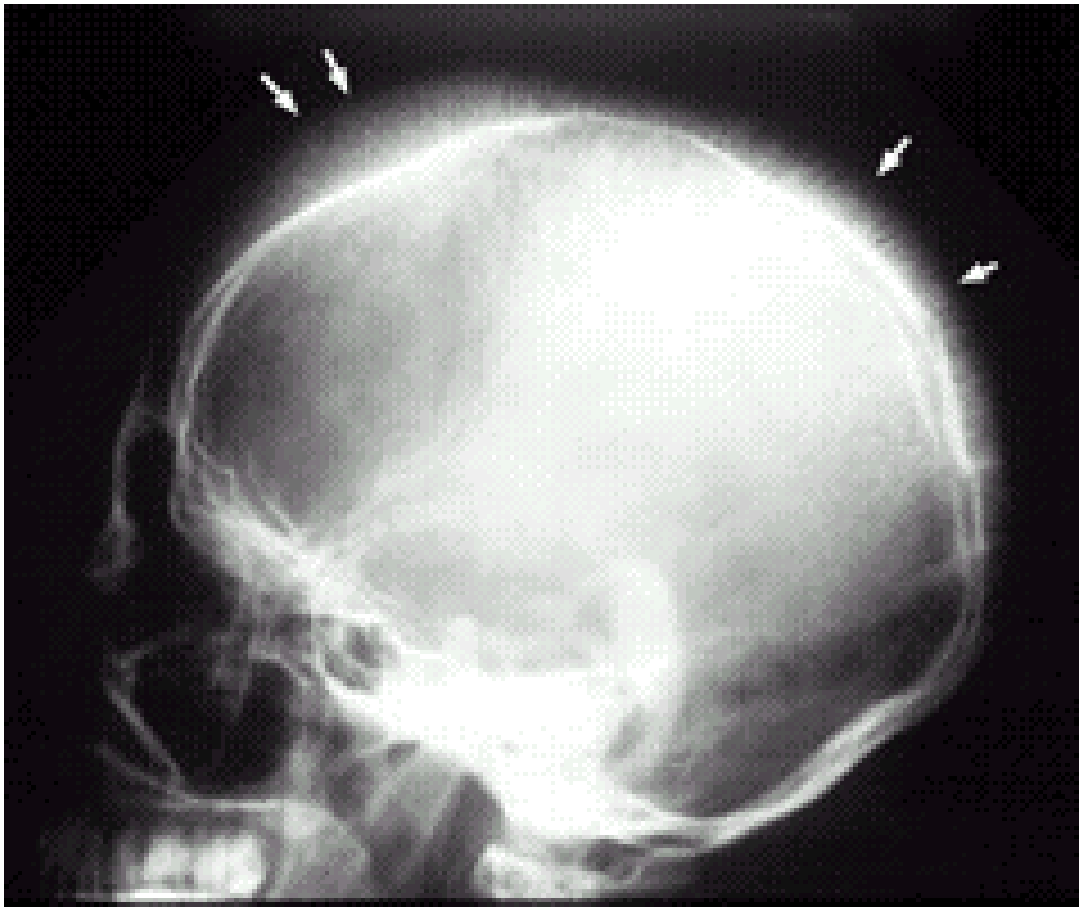
Hepatosplenomegaly



Dark skin due to iron overload



X-Ray Skull
(Hair on End Appearance)



:Thalassemia

- Diagnosis:
  - Hemoglobin Electrophoresis:
    - β-thalassemia minor: ↑ HbA2.
    - β-thalassemia major: ↑ HbF.
    - α-thalassemia:
      - Hb electrophoresis is not diagnostic.
      - DNA analysis with α-gene probes.

Hemolytic Anemia

G6PD Deficiency Anemia

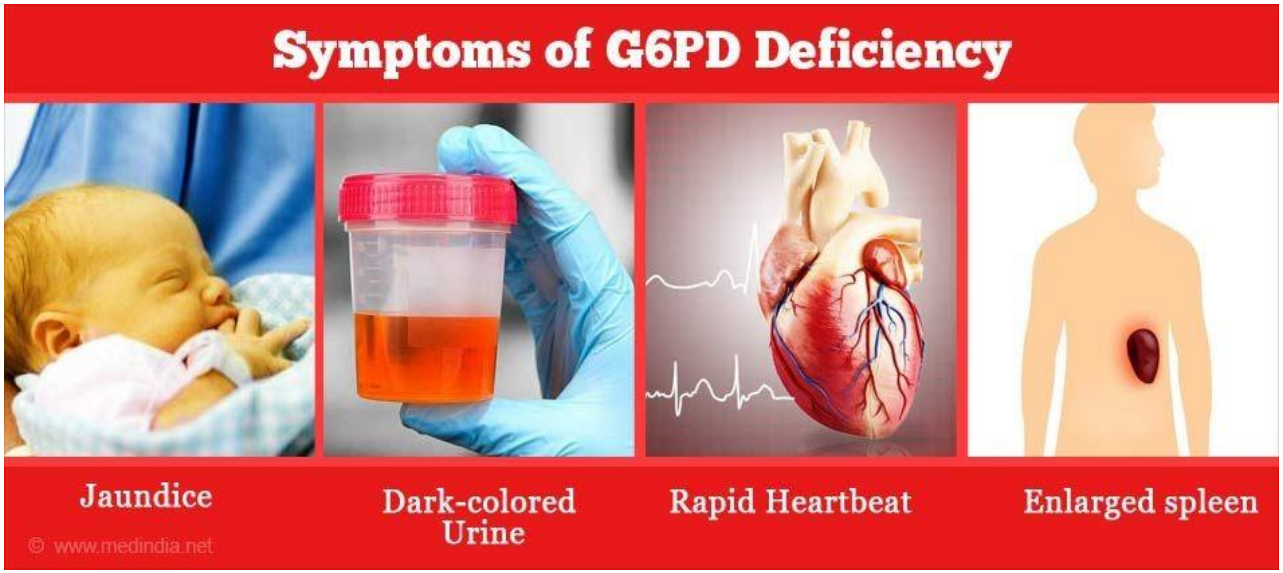
:G6PD Deficiency Anemia

- deficiency in glucose-6-phosphate dehydrogenase (G6PO) leads to a sensitivity of RBC to oxidative stress due to a lack of reduced glutathione.

- X-linked recessive.

:G6PD Deficiency Anemia

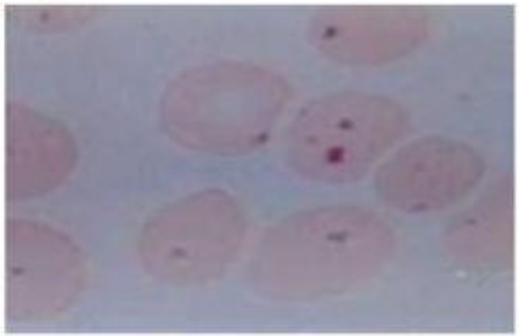
- Clinical Features
  - Frequently presents as episodic hemolysis precipitated by: oxidative stress:
    - drugs (e.g. sulfonamide, antimalarials, nitrofurantoin)
    - infection
    - food (fava beans)



:G6PD Deficiency Anemia

- Investigations:
- **G6PD assay.**
  - should not be done in acute crisis when reticulocyte count is high since reticulocytes have high G6PD levels.

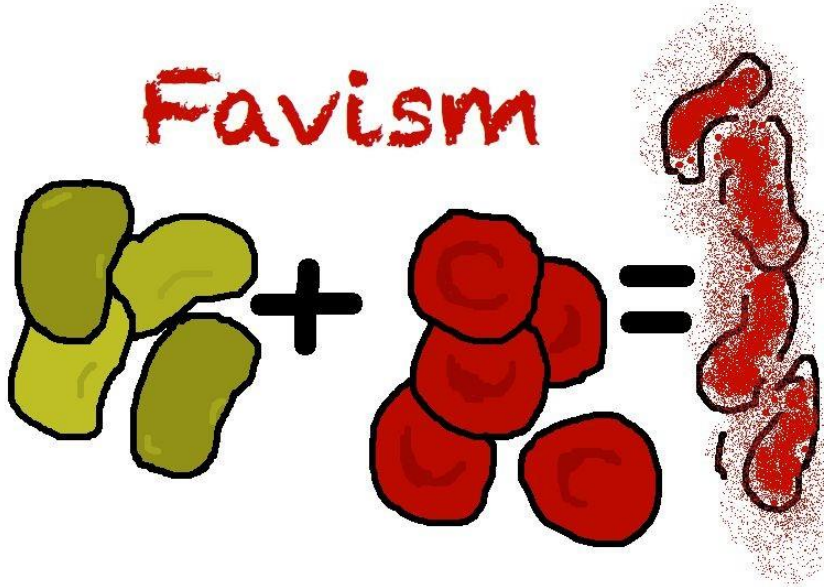
- **Blood film:**
  - Heinz bodies (granules in RBCs due to oxidized Hb); passage through spleen results in the generation of bite cells.



Diagnosis

1- Evidences of haemolysis

2- Enzyme level



Treatment

1 Prevention

2Treatment of acute attack

3- Packed RBCs transfusion

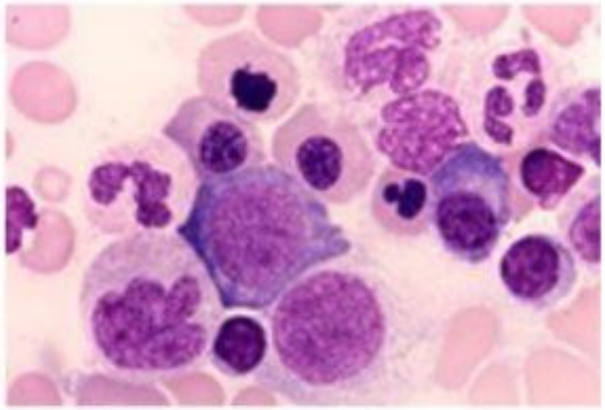
MACROCYTIC ANEMIA

:MACROCYTIC ANEMIA

MEGALOBLASTIC	NON-MEGALOBLASTIC:
□Vitamin B12 def. □Folate def. □Drugs (Methotrexate, Azathioprine).	□Chronic Diseases (2/3). □Reticulocytosis. □MDS. □Liver disease. □Alcohols. □Hypothyroidism.

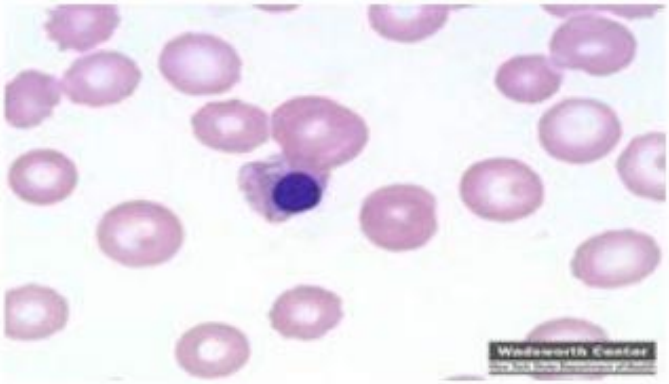
:MACROCYTIC ANEMIA

MEGALOBLASTIC



- Large, oval nucleated RBCs.
- Hypersegmented neutrophil.

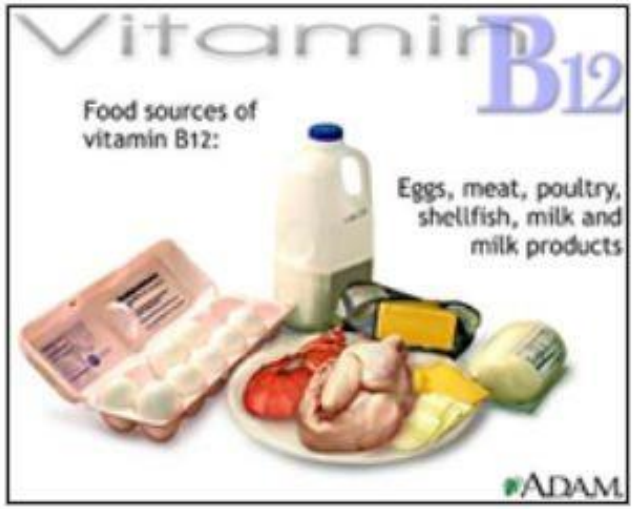
NON-MEGALOBLASTIC:



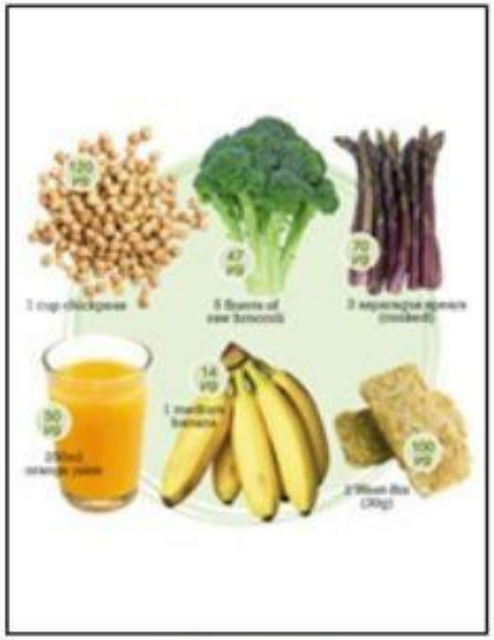
- Large round RBCs.
- Normal neutrophil.

:MEGALOBLASTIC ANEMIA

- Causes:
  - Dietry insufficiency:



Vitamin B12



Folic Acid

- Goat milk: (children) lacks of folic acid.

:MEGALOBLASTIC ANEMIA

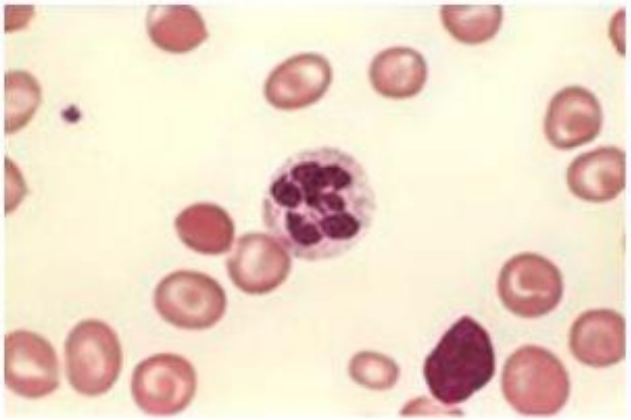
- Causes:
  - Malabsorption:
  - Drugs:
    - Chemotherapy.
    - Anti-epileptics: interfere with folate absorption.
  - Pernicious anemia:
    - Due to destruction of parietal cells, which produce the intrinsic factor needed for vitamin B12 absorption.

:MEGALOBLASTIC ANEMIA

- Clinical Features:
  - fatigue, pallor, diarrhea, loss of appetite, headaches, and tingling/numbness of the hands and feet.
  - Vitamin B12 deficiency affects the nervous system…

:MEGALOBLASTIC ANEMIA

- Investigations:
  - Blood film:
    - Megaloblasts.
    - Hypersegmented neutrophil.

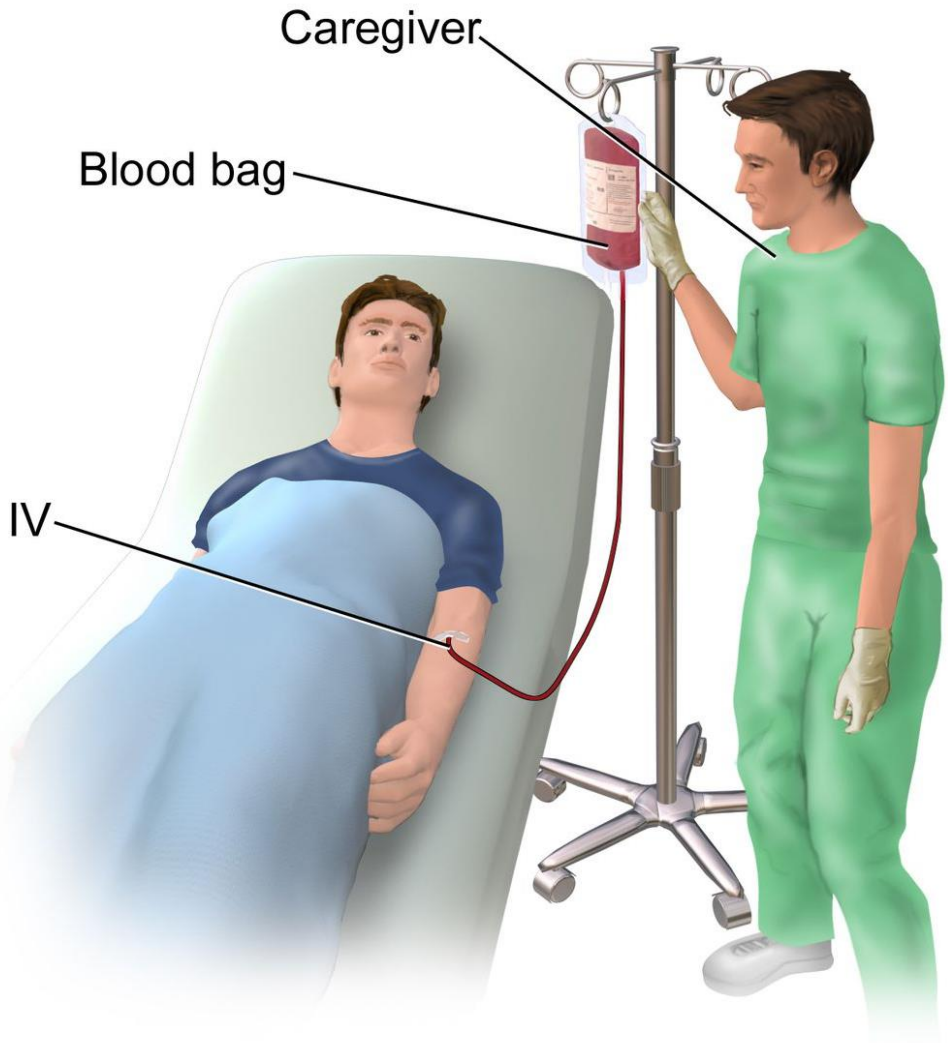


  - Serum vitamin B12 level and RBCs folate.

:MEGALOBLASTIC ANEMIA

- Management:
  - Treat the underlying cause.
  - Vitamin supplements:
  - Improve diet.

Blood Transfusion



Blood Transfusion

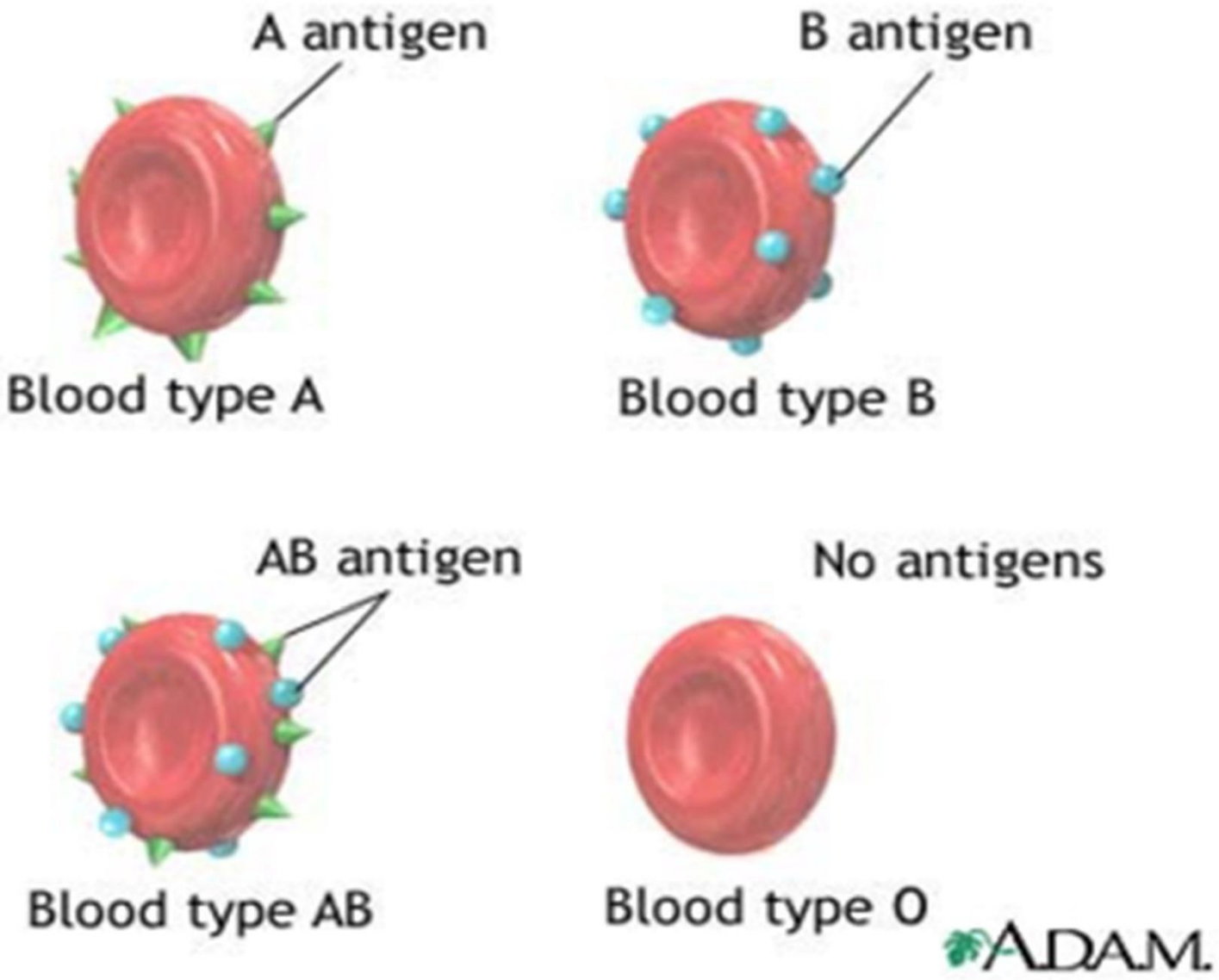
What are the dif erent blood groups?

- There are more than **20 genetically** determined blood group systems known today

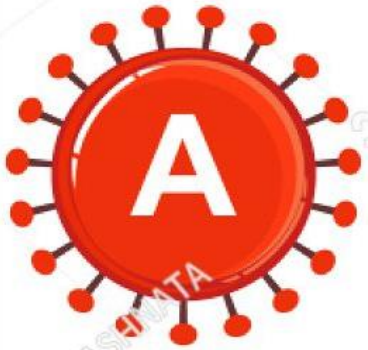
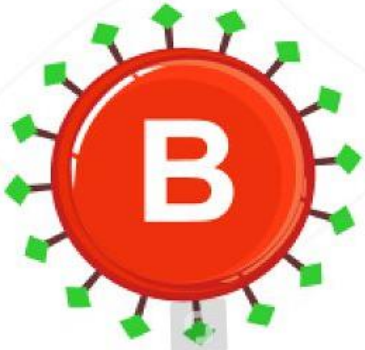
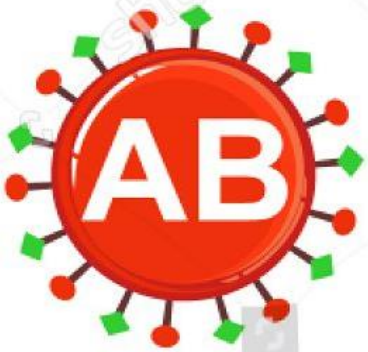
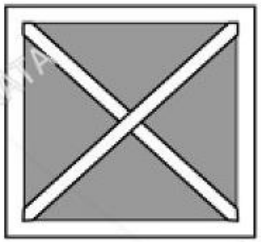
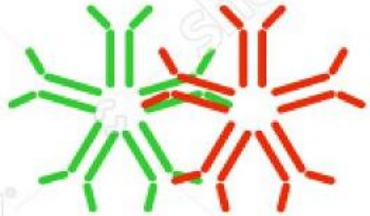
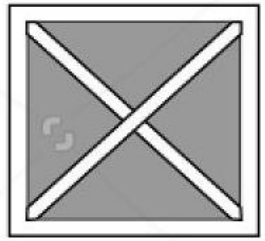
- The **AB0** and **Rhesus (Rh)** systems are the most important ones used for blood transfusions.

- Not all blood groups are compatible with each other.
- Mixing incompatible blood groups leads to blood clumping or agglutination, which is dangerous for individuals.

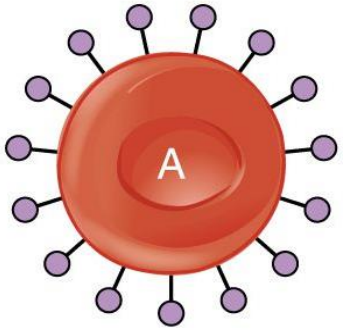
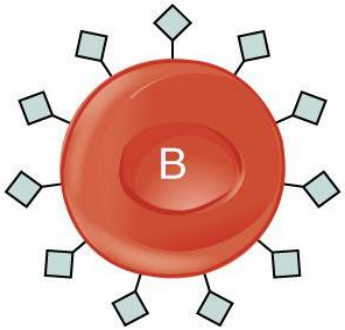
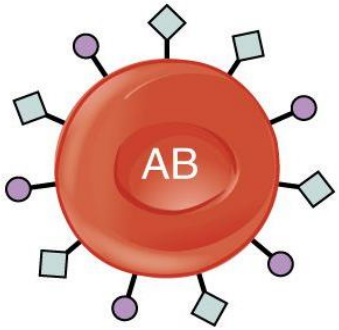
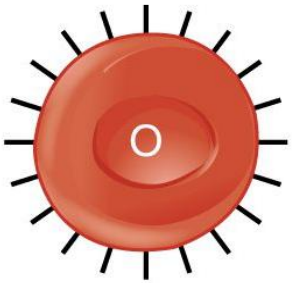
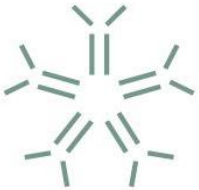
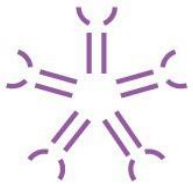
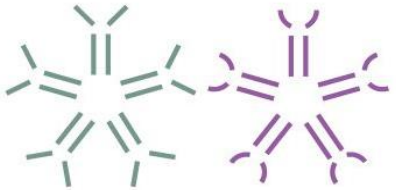
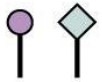
BLOOD TYPES


A antigen
B antigen
Blood type A
Blood type B
AB antigen
No antigens
Blood type AB
Blood type O
A.D.A.M.

CLASSIFICATION OF BLOOD GROUPS

BLOOD TYPE	A	B	AB	O
ERYTHROCYTES				
ANTI BODIES	 anti b	 anti a		 anti b and a
ANTI GENES	 a	 b	 a and b	

Blood Type

	A	B	AB	O
Red Blood Cell Type				
Antibodies in Plasma	 Anti-B	 Anti-A	None	 Anti-A and Anti-B
Antigens in Red blood Cell	 A antigen	 B antigen	 A and B antigens	None
Blood Types Compatible in an Emergency	A, O	B, O	A, B, AB, O (AB^+ is the universal recipient)	O (O is the universal donor)

Possible Blood group Genotypes

Parent Allele	A	B	O
A	AA	AB	AO
B	AB	BB	BO
O	AO	BO	OO

The ABO blood groups

• The most important thing is in assuring **a safe blood transfusion**.

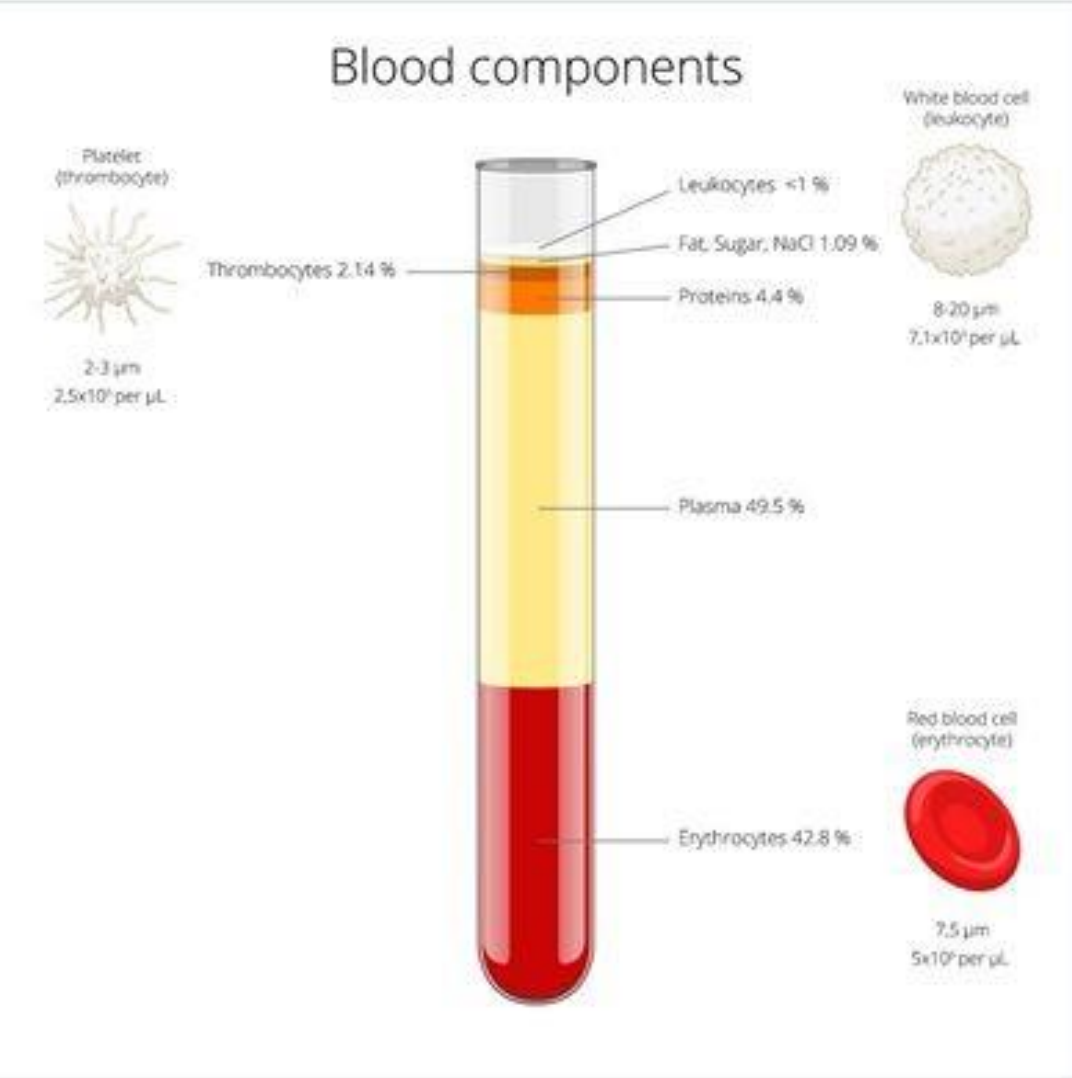
•The table shows the four ABO phenotypes ("blood groups") present in the human population and the genotypes that give rise to them.

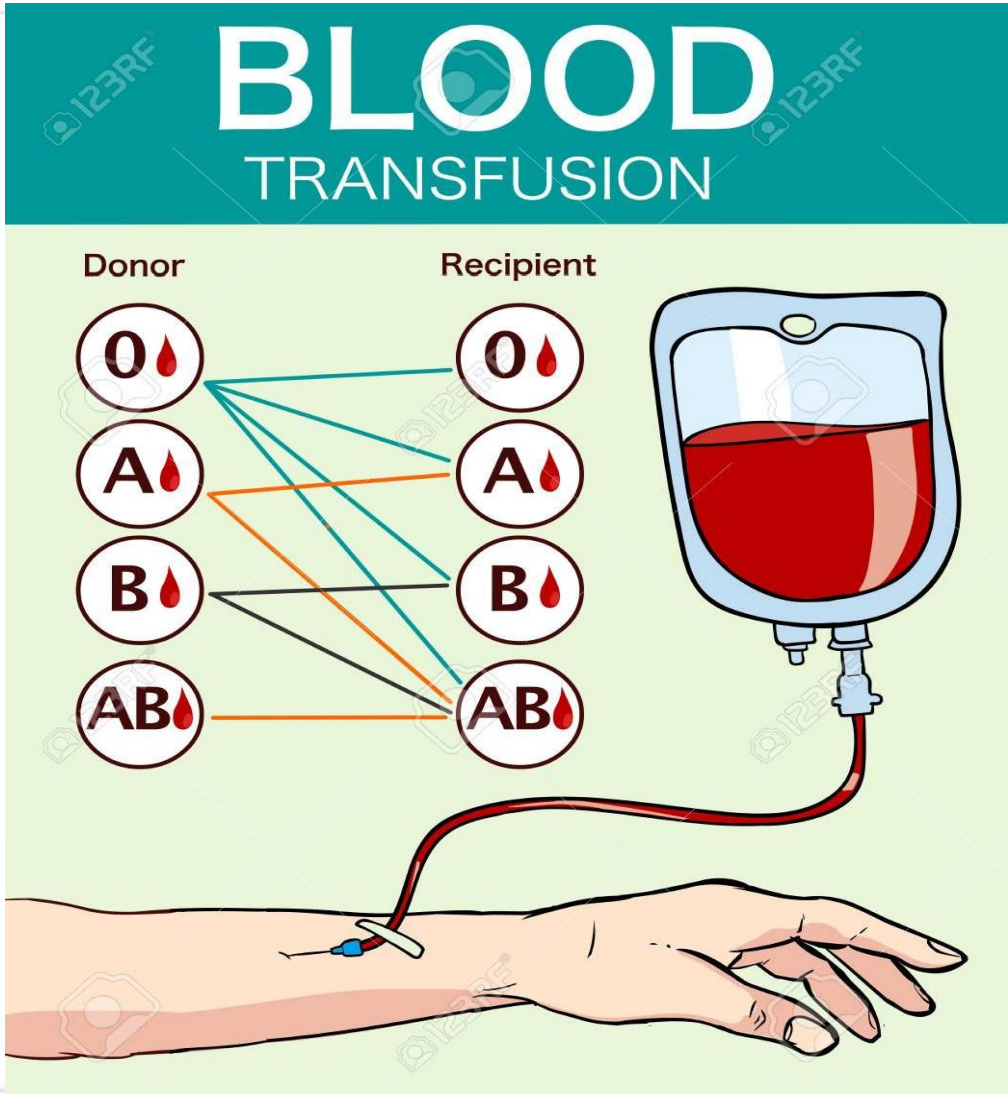
Blood Group	Antigens on RBCs	Antibodies in Serum	Genotypes
A	**A**	Anti-B	*AA* or *AO*
B	**B**	Anti-A	*BB* or *BO*
AB	**A** and **B**	Neither	*AB*
O	Neither	Anti-A and anti-B	*OO*

BLOOD STORAGE

- Blood products must be stored at 4C ± 2C.
- Stored blood has a shelf life of 3 weeks.
- After a storage time of 24-72 hr RBCs have reduced capability to release oxygen to tissues.
- If the patient needs massive transfusions its better to give blood that's less than 7 days old.

- Advantages of component separation
  - Allow optimum survival of each component
  - Transfuse only component needed





- Transfusion practice
  - Transfusion requires doctor's prescription
  - All components MUST be administered through a filter
  - Infuse quickly, within 4 hours
  - D (Rh) -ve require D -ve cellular products
  - ABO identical preferred, ABO compatible OK
  - "Universal donor" – RBCs group O, plasma AB

PRE-TRANSFUSION RESPONSIBILITIES



Blood Transfusion

Type of Transfusion：

- **Whole Blood；**
- **Blood Component；**

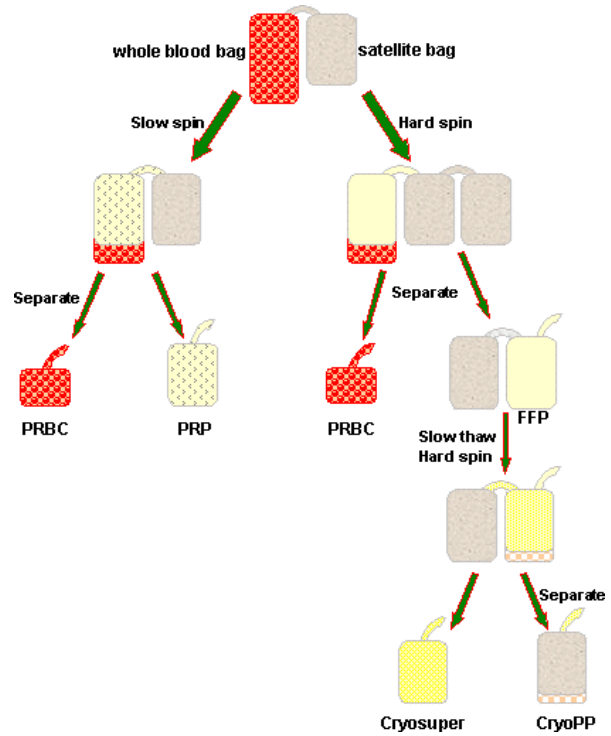
  RBC　PLT　FFP　Leukocyte concentrate

- **Plasma Substitutes；**



Use of whole blood is considered to be a waste of resources

BLOOD COMPONENTS



Whole Blood

Slow Centrifugation

Packed Red Blood Cells

Platelet Rich Plasma

High Speed Centrifugation

1 Unit of Random Donor Platelets

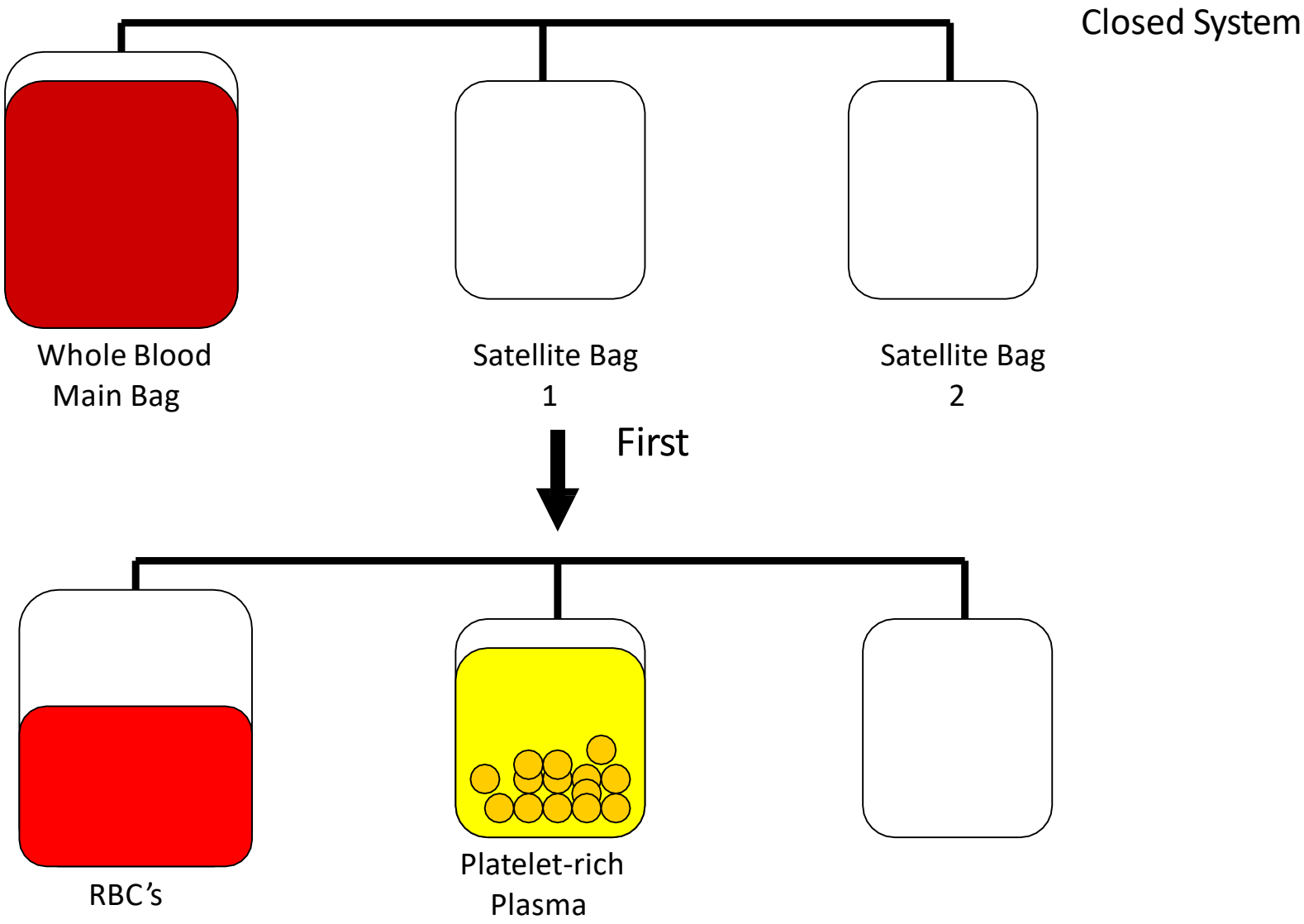
1 Unit of Fresh Frozen Plasma

Thawing precipitates the plasma proteins

Cryoprecipitate

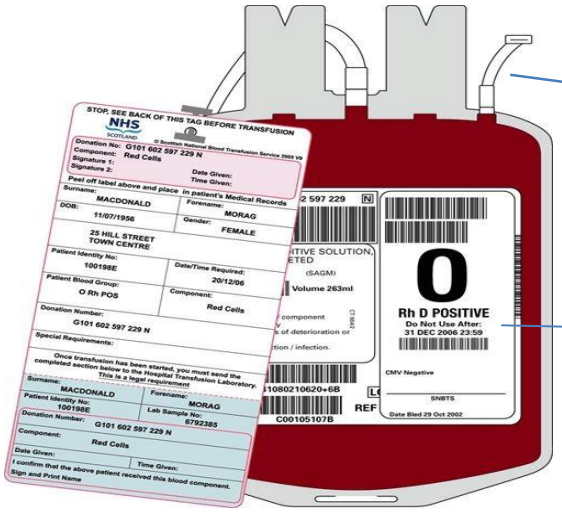
Differential Centrifugation

First Centrifugation



Pre-administration Procedure

- **Step 1:** Check the blood component has been prescribed
- **Step 2:** Undertake baseline observations
- **Step 3:** Undertake visual inspection



Monitoring of Patient

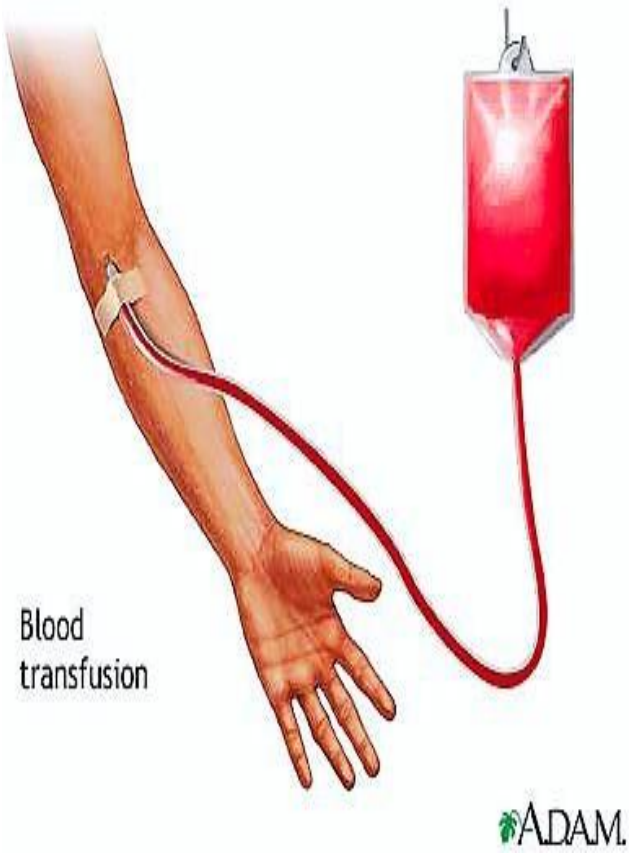
- Base line observations – Temperature, pulse and blood pressure
- Further observations (as above) at 15 minutes
- A set of observations at the end of transfusion
- More frequently if the patient is unwell, unobservable, or unconscious.

Good Documentation

Minimum Transfusion Dataset: the following should be documented in the notes

- Reason for transfusion
- Current blood results
- Component type and amount to be prescribed
- Anticipated outcome
- Any reported transfusion adverse events/reactions
- Review following the transfusion including how much blood has been transfused

PURPOSE OF BLOOD TRANSFUSION THERAPY



* REPLACEMENT
* THERAPEUTIC

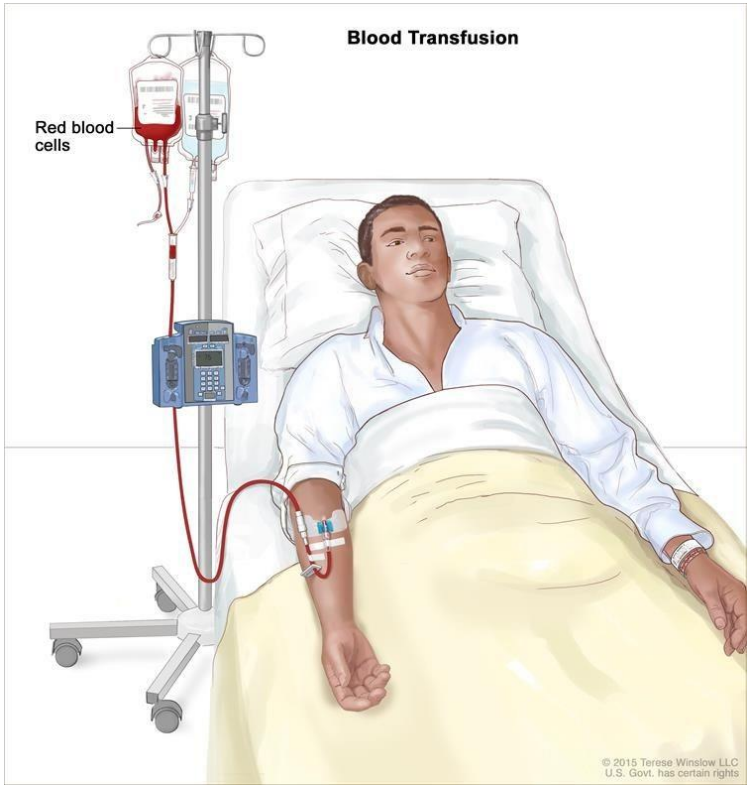
1.To restore intravascular volume with whole blood or albumin.
2.To restore the oxygen capacity of blood by replacing red blood cells.
3.To replace clotting factor and correction of anemia

Types of Blood Transfusion

- Based on time of transfusion
  - Fresh whole blood transfusion
  - Stored Blood

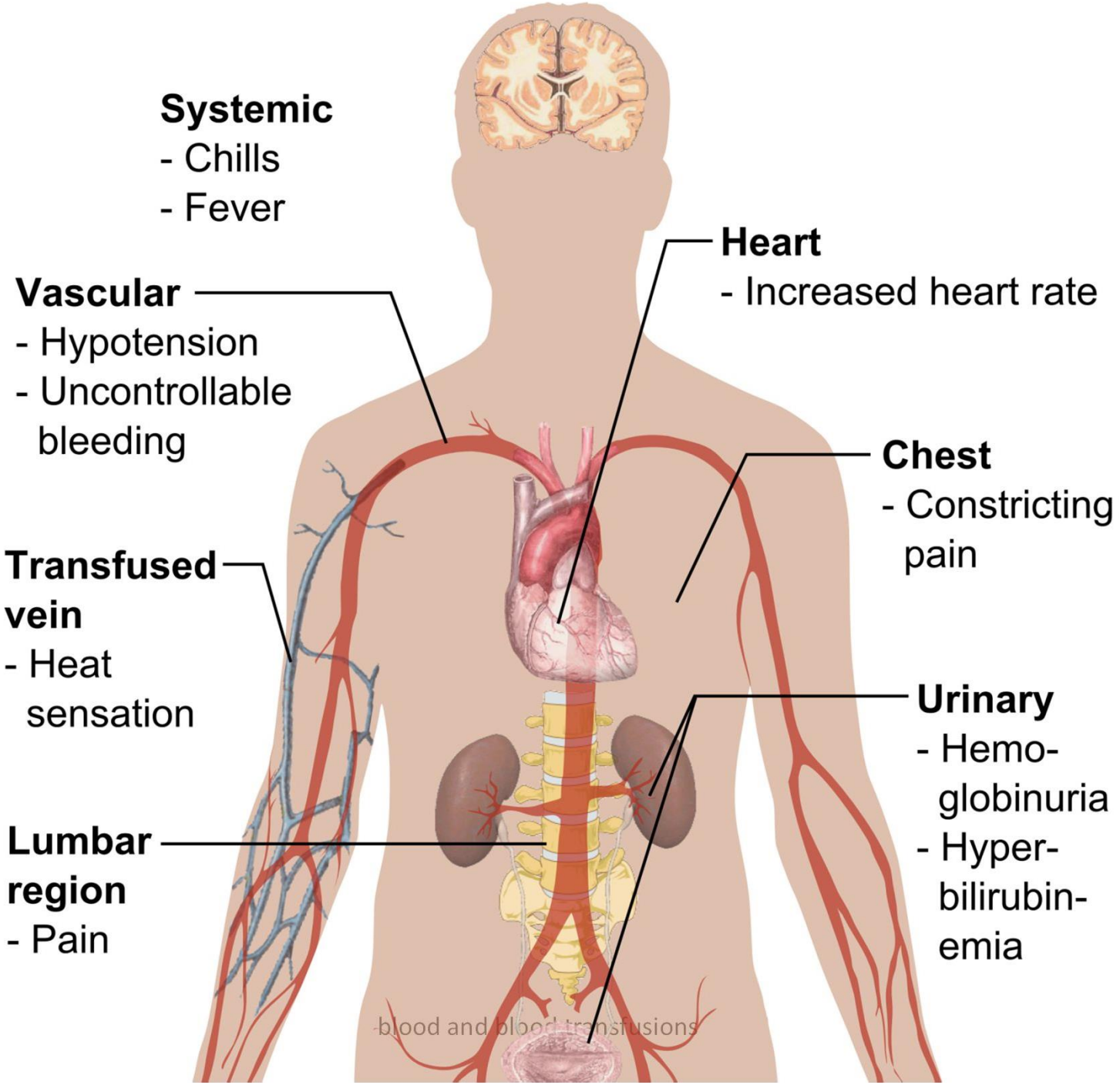
- Based on composition
  - Whole blood
  - Blood fraction

- Based on the donor
  - Autologous: Same donor
  - Blood from different donor



Main symptoms of

Acute hemolytic reaction



Observing / Monitoring the Patient During a Blood / Blood Component Transfusion is part of safe transfusion



- Pyrexia
- Tachycardia
- Hyper / Hypotension
- Headache
- Rigors
- Urticaria - Itchy rash
- Haemoglobinuria
- Collapse
- Nausea / vomiting
- Chest, abdominal, muscle, bone or loin pain
- Generally feeling unwell
- Breathlessness / coughing
- Restlessness Agitation Confusion
- Flushing


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Thank You
www.rayig.8m.com